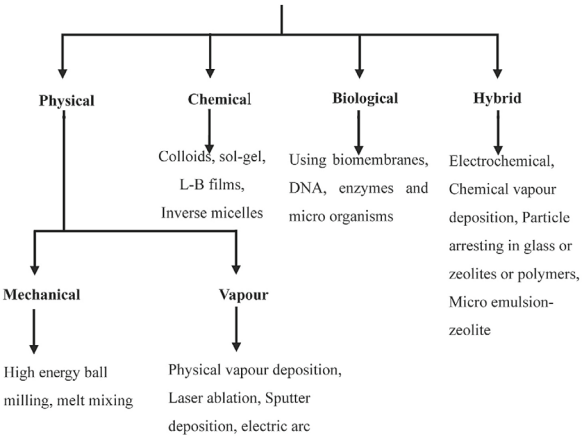
Box 3.1



When the containers are rotating around the central axis as well as their own axis, the material is forced to the walls and is pressed against the walls as illustrated in Fig. 3.2. By controlling the speed of rotation of the central axis and container as well as duration of milling it is possible to ground the material to fine powder (few nm to few tens of nm) whose size can be quite uniform.

Some of the materials like Co, Cr, W, Ni-Ti, Al-Fe and Ag-Fe are made nanocrystalline using ball mill. Few milligrams to several kilograms of nanoparticles can be synthesized in a short time of a few minutes to a few hours.

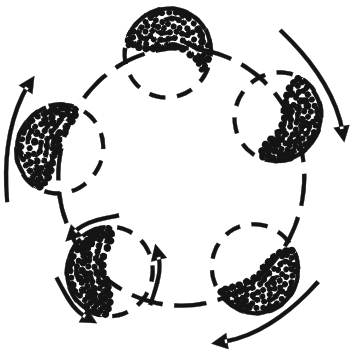


Fig. 3.2: Ball mill in planetary motion (schematic). Sketch shows that the material is thrown against the wall during the course of rotation of a single container. Dark regions are illustrating the powder material, while the rest is empty.

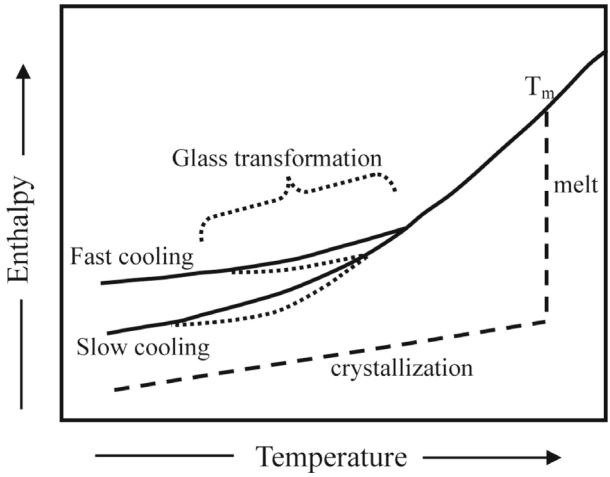


Fig. 3.3: Cooling pattern of glass forming melt.

3.2.2 Melt Mixing

It is possible to form or arrest the nanoparticles in glass. Structurally, glass is an amorphous solid, lacking large range periodic arrangement as well as symmetry arrangement of atoms/molecules. When a liquid is cooled below certain temperature (T_m), as illustrated in Fig. 3.3, it forms either a crystalline or amorphous solid (glass). Besides temperature, rate of cooling and tendency to nucleate decide whether the melt can be cooled as a glass or crystalline solid with long range order. Nuclei are formed spontaneously with homogeneous (in the melt) or inhomogeneous (on surface of other materials) nucleation which can grow to form ordered, crystalline solid. Metals usually form crystalline solids but if cooled at very high rate (~10^5–10^6 K/s), they can form amorphous solids. Such solids are known as metallic glasses. Even in such cases the atoms try to reorganize themselves into crystalline solids. Addition of elements like B, P, Si etc. helps to keep the metallic glasses in amorphous state. It is indeed possible to form nanocrystals within metallic glasses by controlled heating.

Box 3.2
Glass Matrix for Nanoparticle Formation

Although formation of nuclei is necessary for the onset of crystallization, presence of large number of nuclei of different elements creates an inhomogeneous melt favouring glass formation. There are some materials that particularly favour glass formation. Silicates (or germanates) have a central silicon atom in a pyramidal structure with oxygen atoms at the corners as shown in Fig. 3.4. Such silicate units, not sharing the edges, but connected only through the corners are randomly distributed in 3-D

to form glassy structure. A window glass has in addition to SiO_2 (~72%), oxides of sodium (~14.5%), calcium (~8.5%), magnesium (~3.5%) and aluminium (~1.5%) as its constituents. Laboratory glassware has silica (80%), boron oxide (10%), sodium oxide (5%), alumina (3%), magnesium oxide (1%) and calcium oxide (1%). Other glasses like fibre glass and quartz glass have different compositions. It can be however noticed that glass formation is largely assisted by the presence of number of different elements (heterogeneous nature). It has been found that beautiful colours in glasses like red, yellow, blue, orange, green and their shades are a result of addition of some elements like gold, cobalt, nickel, iron etc. These colours are attributed to the formation of nanoparticles of these elements. Indeed it is also possible to form nanoparticles of some semiconductors like CdS, CdSe, ZnS and TiO_2 in glass. Usually the percentage of such additives is as small as <5%. The melt of glass forming materials and desired nanoparticle material is well homogenized before the cooling begins. The nanoparticles are usually well separated and immobilized by glassy matrix.

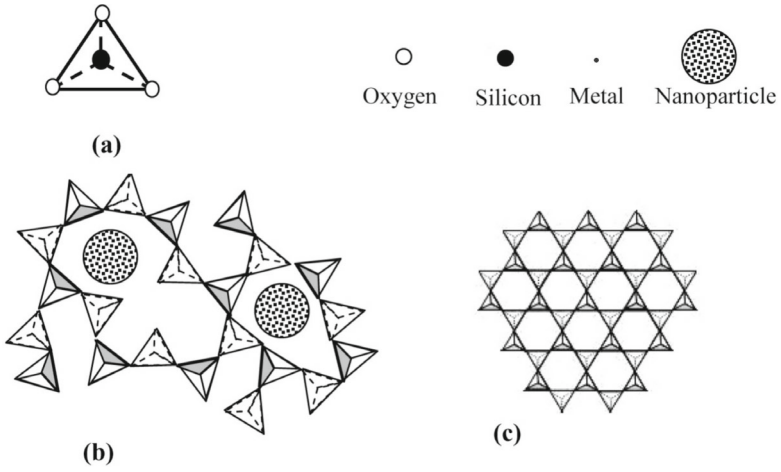


Fig. 3.4: Structure of silicates. (a) Tetrahedral (pyramidal) building block of typical silicates or germanets. (b) Disordered arrangement of tetrahedral units to form a 'glassy' substance. This entraps nanoparticles inside appropriate voids. (c) Tetrahedron can also form crystalline regular arrangement.

It is also possible to form some nanoparticles by mixing the molten streams of metals at high velocity with turbulence. On mixing thoroughly, nanoparticles are found to have been formed. For example a molten stream of Cu-B and molten stream of Ti form nanoparticles of TiB_2.

3.3 METHODS BASED ON EVAPORATION

There is a variety of methods to form nanostructures by evaporating the materials on some substrates. The nanostructured materials can be in the form of thin films, multilayer films or nanoparticulate thin films (thin films composed of nanoparticles). There are several methods in which material of interest is brought in the gaseous phase (atoms or molecules) which can form clusters and then deposit on appropriate substrates. It is also possible to obtain very thin (even atomic layers, known as monolayers) layers or multilayers (multilayers are layers of two or more materials stacked over each other) forming nanomaterials of wide interest.

Evaporation can be achieved by various methods like resistive heating, electron beam heating, laser heating, sputtering. It should be remembered that all the synthesis processes need to be carried out in a properly designed vacuum system (see Appendix IV for vacuum techniques), so as to avoid uncontrolled oxidation of source materials and final product as well as that of components of the synthesis system. Mean free path of the particles also increases in vacuum system, which is often desired. Even if some reactive gases are used in certain cases of depositions, it is useful first to evacuate the system to very low pressure so that the materials to be synthesized do not get contaminated by undesired atoms and then pressurize the system to desired value by introducing the high purity gases in the synthesis chamber.

Materials to be evaporated are usually heated from some suitable filament, crucible, boat (collectively called as ‘evaporation source’ or ‘crucible’) as shown in Fig. 3.5.

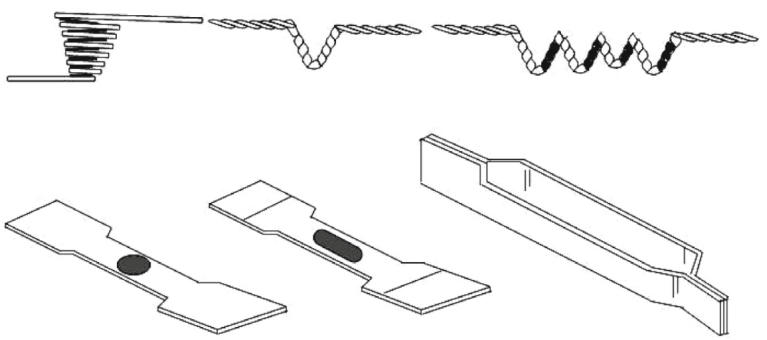


Fig. 3.5: Different shapes of filaments, canoe and baskets used for holding the materials for evaporation.

Usually the sources are electrically heated so that enough vapours of the material to be deposited are generated. If the material to be deposited wets the filament material without forming any alloy or compound, the filament is considered to be suitable. Otherwise one needs to melt the material in a basket, canoe-like container. However this type of heating has the disadvantage that the crucible itself and surrounding parts also get heated and become the source of unwanted contamination or impurities. Therefore evaporation by electron beam heating method is desired. Electron beam focuses on the material to be deposited, kept in the crucible as it is generated from a filament that is not in the proximity of the evaporating material. It melts only some central portion of the material in crucible avoiding any contamination from crucible. Thus high purity vapours of materials can be obtained.

Now, let us discuss how the deposition occurs by evaporation. At any given temperature there is some vapour pressure of the material. In evaporation, the number of atoms leaving the surface of solid or liquid material should exceed the atoms returning to the surface. Evaporation rate from a liquid is given by Hertz-Knudsen equation as follows

$$\frac{dN}{dt} = A\alpha(2\pi mkT)^{-1/2}(p^* - p) \tag{3.1}$$

where N is the number of molecules or atoms leaving a liquid or solid surface, A – area from which atoms evaporate, α – coefficient of evaporation, m – mass of evaporating atoms, k – Boltzman constant, T – temperature, p^* – equilibrium pressure at the source of evaporation and p is hydrostatic pressure on the surface.

If none of the evaporated molecules or atoms return to the surface of the evaporation source, $p = 0$. The coefficient of evaporation arises as some of the atoms/molecules returning to the surface get reflected back into vapour phase and also change the pressure due to evaporant. Equation (3.1) also suggests that at a given temperature, there would be some specific rate of deposition. It may be noted that evaporation rate equations would in general be more complicated than given by a simple Hertz-Knudsen equation when evaporation from solids, compounds, alloys etc. are taken into account.

It is necessary that the material to be evaporated creates a pressure of ~10^{-1} Pa or more, so as to achieve adequate vapour pressure for synthesis. Some materials like Ti, Mo, Fe and Si have large vapour pressure at a temperature, much below their melting points and can be easily evaporated or sublimated from their solids. On the other hand metals like Au and Ag have very low vapour pressures even close to their melting points. Therefore, they need to be melted (for evaporation to occur) to achieve adequate vapour pressure required for deposition.

The constituents of the alloys evaporate at different rates depending upon their pure metal forms. Therefore, the deposited films may have different stoichiometry (% amounts of different constituents in a material) as compared to the original alloy composition.

If a compound is used for evaporation, it is very likely that it would dissociate beforehand and stoichiometry is not maintained in the final product.

Use of reactive gases like oxygen, ammonia and hydrogen during the material synthesis by evaporation is a common practice. Sometimes post-deposition treatments are also used.

In fact in a situation where the material to be deposited is not a single component material, it is preferable to use sputter deposition.

3.3.1 Physical Vapour Deposition with Consolidation

This technique basically involves use of materials of interest as sources of evaporation, an inert gas or reactive gas for collisions with material vapour, a cold finger on which clusters or nanoparticles can condense, a scraper to scrape the nanoparticles and piston-anvil (an arrangement in which nanoparticle powder can be compacted). All the processes are carried out in a vacuum chamber so that the desired purity of the end product can be obtained. Figure 3.6 schematically illustrates a set up for carrying out physical vapour deposition and compressing the powder in a pellet form.

Usually metals or high vapour pressure metal oxides are evaporated or sublimated from filaments or boats of refractory metals like W, Ta and Mo in which the materials to be evaporated are held. The density of the evaporated material close to the source is quite high and particle size is small (<5 nm). Such particles would prefer to acquire a stable lower surface energy state. Due to small particle or cluster–cluster interaction bigger particles get formed. Therefore, they should be removed away as fast as possible from the source. This is done by forcing an inert gas near the source, which removes the particles from the vicinity of the source. In general the rate of

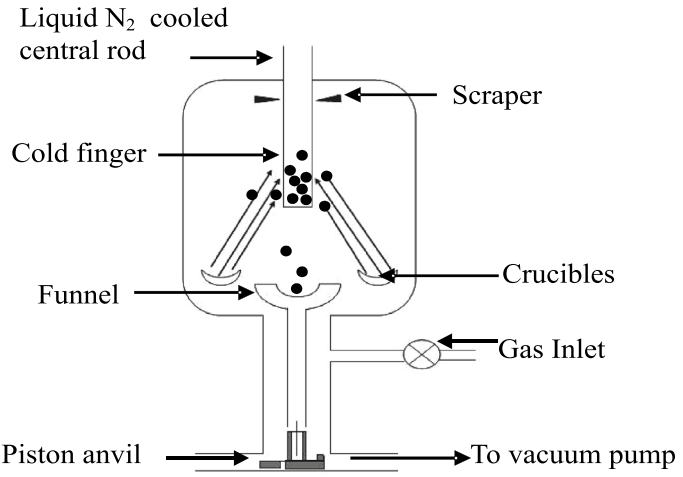


Fig. 3.6: Schematic diagram for the synthesis of nanoparticles by physical vapour deposition. Particles condense on the central cooled rod, are scraped and consolidated using a piston-anvil system in vacuum to form a pellet.

evaporation and the pressure of gases inside the chamber determine the particle size and their distribution. Distance of the source from the cold finger is also important. Evaporated atoms and clusters tend to collide with gas molecules and make bigger particles, which condense on cold finger. While moving away from the source to cold finger the clusters grow. If clusters have been formed on inert gas molecules or atoms, on reaching the cold finger, gas atoms or molecules may leave the particles there and then escape to the gas phase. If reactive gases like O_2, H_2 and NH_3 are used in the system, evaporated material can interact with these gases forming oxide, nitride or hydride particles. Alternatively one can first make metal nanoparticles and later make appropriate post-treatment to achieve desired metal compound etc. Size, shape and even the phase of the evaporated material can depend upon the gas pressure in deposition chamber. For example using gas pressure of H_2 more than 500 kPa, TiH_2 particles of ~12 nm size were produced. By annealing them in O_2 atmosphere, they could be converted into titania (TiO_2) having rutile phase. However if titanium nanoparticles were produced in H_2 gas pressure less than 500 kPa, they could not be converted into any crystalline oxide phase of titanium but always remained amorphous.

Clusters or nanoparticles condensed on the cold finger (water or liquid nitrogen cooled) can be scraped off inside the vacuum system. The process of evaporation and condensation can be repeated several times until enough quantity of the material falls through a funnel in which a piston-anvil arrangement has been provided. One can even have separate low and high pressure presses. A pressure of few mega pascal (MPa) to giga pascal (GPa) is usually applied depending upon the material. Low porosity pellets are easily obtained. Density of the material thus can be ~70 to 90% of the bulk material.

3.3.2 Ionized Cluster Beam Deposition

This method was first developed by Takagi and Yamada around 1985 and is also useful to obtain adherent and high quality single crystalline thin films. The set up consists of a source of evaporation, a nozzle through which material can expand into the chamber, an electron beam to ionize the clusters, an arrangement to accelerate the clusters and a substrate on which nanoparticle film can be deposited, all housed in a suitable vacuum chamber. A schematic arrangement is illustrated in Fig. 3.7.

Small clusters from molten material are expanded through the fine nozzle. The vapour pressure ~1 Pa to 1 kPa needs to be created in the source and the nozzle needs to have a diameter larger than the mean free path of atoms or molecules in vapour form in the source to form the clusters. On collision with electron beam, clusters get ionized. Due to applied accelerating voltage, the clusters are directed towards the substrate. By controlling the accelerating voltage, it is possible to control the energy

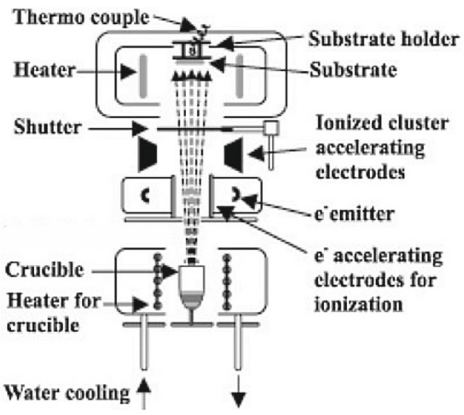


Fig. 3.7: Ionized cluster beam apparatus.

with which the clusters hit the substrate. Stable clusters of some materials would require considerable energy to break their bonds and would rather prefer to remain as small clusters of particles. Thus it is possible to obtain the films of nanocrystalline material using ionized cluster beam. However it is not unlikely that some neutral atoms also get incorporated in the film. Besides, the clusters are not mass selected. Therefore they can have wide distribution of particle sizes. In fact anything that can cross the accelerating voltage can get incorporated in the film.

3.3.3 Laser Vapourization (Ablation)

In this method, vapourization of the material is effected using pulses of laser beam of high power. The set up along with the interaction with the evaporation source is depicted in Fig. 3.8.

The set up is an Ultra High Vacuum (UHV) or high vacuum system equipped with inert or reactive gas introduction facility, laser beam, solid target and cooled substrate. Clusters of any material of which solid target can be made are possible to synthesize. Usually laser operating in the UV range such as excimer (excited diamers or exciplex (excited complex)) laser (see Table 3.1) is necessary because other wavelengths like IR or visible are often reflected by surfaces of some metals.

A powerful beam of laser evaporates the atoms from a solid source and atoms collide with inert gas atoms (or reactive gases) and cool on them forming clusters. They condense on the cooled substrate. The method is often known as laser ablation. Gas pressure is very critical in determining the particle size and distribution. Simultaneous evaporation of another material and mixing the two evaporated materials in inert gas leads to the formation of alloys or compounds. This method can produce some novel phases of the materials which are not normally formed. For example Single Wall Carbon Nanotubes (SWNT) are mostly synthesized by this method.

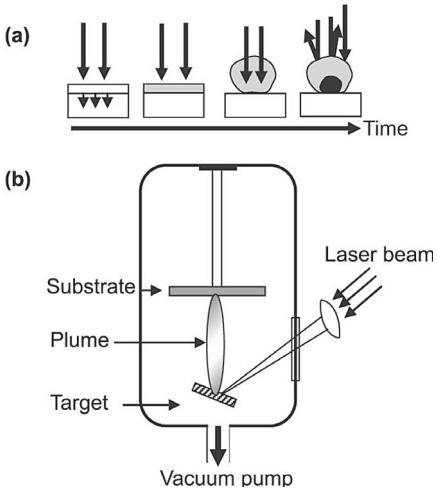


Fig. 3.8: (a) Sequence of material evaporation by laser beam interaction with a target material. (b) Laser deposition schematic apparatus.

Table 3.1: Wavelengths of some commonly used excimer lasers

Gas	F_2	ArF	KrCl	KrF	XeCl	XeF
λ, nm	157	193	222	249	308	350

3.3.4 Laser Pyrolysis

Another method of thin films synthesis using lasers is known as 'laser pyrolysis' or laser-assisted deposition. Here a mixture of reactant gases is decomposed using a powerful laser beam in presence of some inert gas like helium or argon. Atoms or molecules of decomposed reactant gases collide with inert gas atoms and interact with each other, grow and are then deposited on cooled substrate. Schematic diagram is given in Fig. 3.9.

Many nanoparticles of materials like Al_2O_3, WC and Si_3N_4 are synthesized by this method. Here too, gas pressure plays an important role in deciding the particle sizes and their distribution.

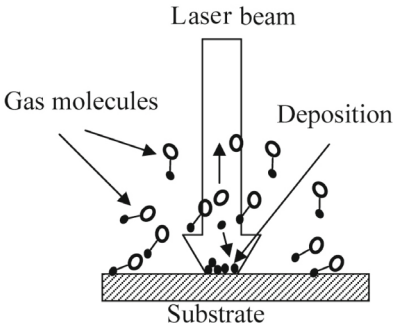


Fig. 3.9: Schematic diagram depicting laser pyrolysis.

3.4 SPUTTER DEPOSITION

Sputter deposition is a widely used thin film deposition technique, specially to obtain stoichiometric thin films (i.e. without changing the composition of the original material) from target material. Target material may be some alloy, ceramic or compound. Sputtering is also effective in producing non-porous compact films. It is a very good technique to deposit multilayer films for mirrors or magnetic films for spintronics applications.

In sputter deposition, some inert gas ions like Ar^+ are incident on a target at a high energy. Depending on the energy of ions, ratio of ion mass to that of target atoms mass, the ion-target interaction can be a complex phenomenon. The ions become neutral at the surface but due to their energy, incident ions may get implanted, get bounced back, create collision cascades in target atoms, displace some of the atoms in the target creating vacancies, interstitials and other defects, desorb some adsorbates, create photons while loosing energy to target atoms or even sputter out some target atoms/molecules, clusters, ions and secondary electrons. Figure 3.10 shows a schematic picture of various possibilities.

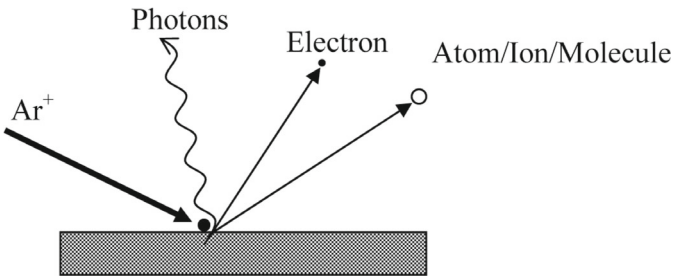


Fig. 3.10: Interaction of an ion with target.

For deposition of materials, one is interested in sputtering out the target material. Target material sputter yield is given by

$$Y = \frac{3\alpha}{4\pi^2} \frac{4M_1M_2}{(M_1 + M_2)^2} \frac{E_1}{E_b} \tag{3.2}$$

for

$$E_1 < 1\,\text{keV}$$

and

$$Y = 3.56\alpha \frac{Z_1Z_2}{Z_1^{2/3} + Z_2^{2/3}} \left(\frac{M_1}{M_1 + M_2} \right) \frac{S_n(E_2)}{E_b} \tag{3.3}$$

for

$$E_2 > 1\,\text{keV}$$

where α is efficiency of momentum transfer, M_1 – mass of incident ion, M_2 – mass of target atom, Z_1 – atomic number of incident ion, Z_2 – atomic number of target atom, E_1 – energy of incident ion, E_b – binding energy of target atom and S_n is known as stopping power. It is an energy loss per unit length due to nuclear collisions.

Box 3.3
Creating Plasma

The terms glow discharge and plasma are often used to mean the same thing. Plasma is a mixture of free electrons, ions and photons. Plasma is overall neutral but there can be regions which are predominantly of positive or negative charges. One can get plasma in different gases at different pressures by using a vacuum tube with two metal electrodes and applying high DC or AC voltage. Different regions of plasma generated by applying a high DC voltage can be divided into five regions as

(i) Cathode or Crook's dark space,
(ii) Negative space glow,
(iii) Faraday's dark space,
(iv) Positive column,
(v) Anode dark space.

The extent of these regions depends upon the pressure of the gas.

The glow discharge in a glass tube with two electrodes at the end filled with air has following characteristics at different pressures (Fig. 3.11).

Pressure (torr)	*Appearance of discharge spark*
~10	General glow discharge
~1–0.5	Closely spaced striations in positive column
~0.5	10 mm spacing of striations
~0.3	Crook's dark space 5 mm long
~0.1	Crook's dark space 10 mm long
~0.05	Crook's dark space 20 mm long
~0.005	Discharge disappearance

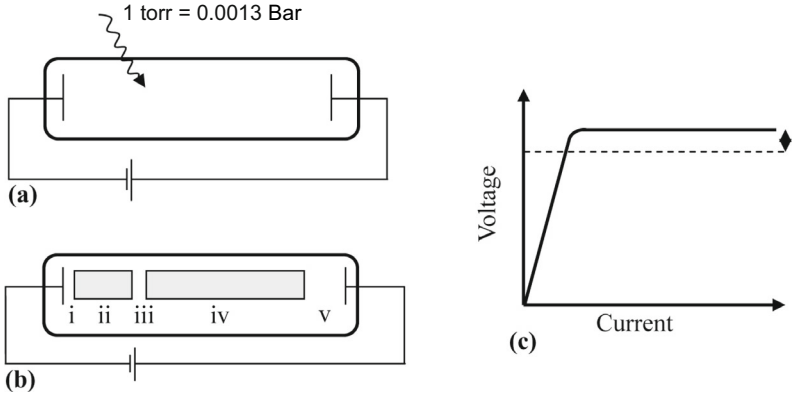


Fig. 3.11: (a) Generation of plasma by applying high voltage between two electrodes in an evacuated glass tube; (b) Different regions of plasma and (c) Typical current–voltage characteristic.

Sputter yield for different elements with same incident ion having same energy varies in general. This would lead one to think that from a target consisting of two different elements or more, the one having higher sputter yield should get incorporated in larger quantity than the others. However high sputter yield elements get depleted fast and other elements also make contribution. Thus the stoichiometry is achieved in the deposited film.

Sputter deposition can be carried out using Direct Current (DC) sputtering, Radio Frequency (RF) sputtering or magnetron sputtering. In all the above cases one can also use discharge or plasma of some inert gas atoms or reactive gases. The deposition is carried out in a high vacuum or ultra high vacuum system equipped with electrodes, one of which is a sputter target and the other is a substrate, gas introduction facility etc. Although the system during deposition is at high gas pressure, low base pressure ensures that the adequate purity is obtained.

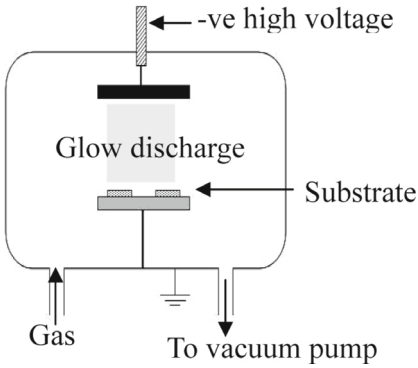


Fig. 3.12: Schematic layout of a typical DC sputtering unit.

3.4.1 DC Sputtering

This is a very straight forward technique of deposition, in which sputter target is held at high negative voltage and substrate may be at positive, ground or floating potential (see Fig. 3.12). Substrates may be simultaneously heated or cooled depending upon the material to be deposited. Once the required base pressure is attained in the vacuum system, usually argon gas is introduced at a pressure <10 Pa. A visible glow is observed and current flows between anode and cathode indicating the deposition onset. When sufficiently high voltage is applied between anode and cathode with a gas in it a glow discharge is set up with different regions as cathode glow, Crooke's dark space, negative glow, Faraday dark space, positive column, anode dark space and anode glow (Box 3.3). These regions are the result of plasma, i.e. a mixture of electrons, ions, neutral atoms and photons released in various collisions. The density of various particles and the length over which they are distributed depends upon the gas pressure. Energetic electron impacts cause gas ionization. Ratio of ions/neutrals can be typically ~10^{-4}. Thus at a few Pa pressure, sufficiently large number of ions are generated that can be used to sputter the target.

3.4.2 RF Sputtering

If the target to be sputtered is insulating, it is difficult to use DC sputtering. This is because it would mean the use of exceptionally high voltage (>10^6 V) to sustain discharge between the electrodes. (In DC discharge sputtering 100 to 3000 V is usual.) However if some high frequency voltage is applied the cathode and anode alternatively keep on changing the polarity and oscillating electrons cause sufficient ionization.

In principle, 5 to 30 MHz frequency can be used and the electrodes can be insulating. However, 13.56 MHz frequency is commonly used for deposition, as this frequency is reserved worldwide for this purpose and many others are available for

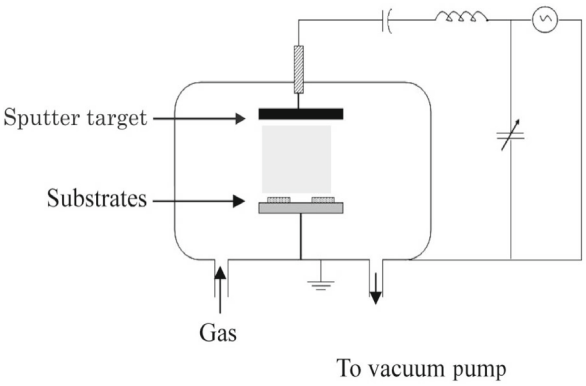


Fig. 3.13: RF sputtering set up with matching unit.

communication. Target itself biases to negative potential becoming cathode when the arrangement as depicted in Fig. 3.13 is used.

3.4.3 Magnetron Sputtering

RF/DC sputtering rates can further be increased by using magnetic field. When both electric and magnetic fields act simultaneously on a charged particle, the force on it is given by Lorentz force.

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{3.4}$$

If **E** and **B** are parallel to each other as illustrated in Fig. 3.14 and an electron leaves the cathode at angle $\theta = 0°$, then $\mathbf{v} \times \mathbf{B} = 0$ and only electric field vector acts on an electron. However an electron making an angle θ would have both electric and magnetic fields acting on it. The magnitude of velocity component in the direction of electric field would be $q\mathbf{v}\cos\theta$. The component due to magnetic field would be perpendicular to both **v** and **B** with a magnitude $qvB\sin\theta$. Electron orbits around the magnetic field axis with radius r. The centripetal force on it would be $m(v\sin\theta)^2/r$

$$\frac{mv^2\sin^2\theta}{r} = qvB\sin\theta \tag{3.5}$$

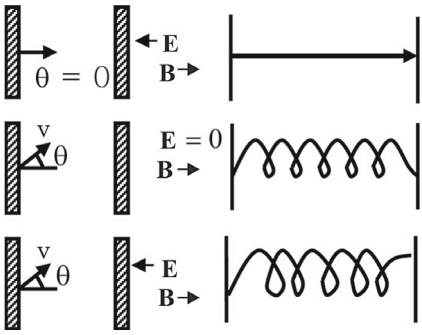


Fig. 3.14: Effect of **E** and **B** on electron.

Thus

$$r = \frac{mv \sin \theta}{qB} \tag{3.6}$$

Electron moves in a helical path and is able to ionize more atoms in the gas. In practice, both parallel and perpendicular magnetic fields to the direction of electric field are used to further increase the ionization of the gas, increasing the efficiency of sputtering. By introducing gases like O_2, N_2, NH_3, CH_4 and H_2S, while metal targets are sputtered, one can obtain metal oxides like Al_2O_3, nitrides like TiN and carbides like WC. This is known as 'reactive sputtering'.

3.4.4 ECR Plasma Deposition

The plasma density can be further enhanced using microwave frequency and coupling the resonance frequency of electrons in magnetic field as

$$\omega = \frac{eB}{m} \tag{3.7}$$

where ω is microwave frequency, often 2.45 GHz, e – charge of electron, m – mass of free electron and B is magnetic field (870 Gauss).

Ionization density using ECR plasma is about 2–3 orders of magnitude larger than that by previous methods. Thin films and nanoparticles of SiO_2, SiN and GaN have been obtained by using this technique. Schematic of this technique is shown in Fig. 3.15.

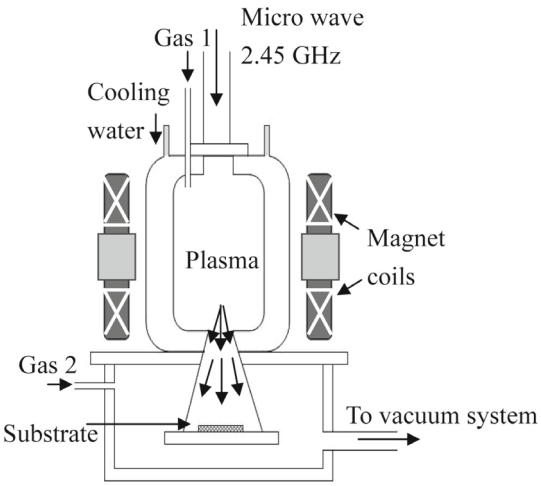


Fig. 3.15: Schematic diagram for Electron Cyclotron Resonance (ECR) for deposition.

3.5 CHEMICAL VAPOUR DEPOSITION (CVD)

Chemical vapour deposition, a hybrid method using chemicals in vapour phase is conventionally used to obtain coatings of a variety of inorganic or organic materials. It is widely used in industry because of relatively simple instrumentation, ease of processing, possibility of depositing different types of materials and economical viability. Under certain deposition conditions nanocrystalline films or single crystalline films are possible. There are many variants of CVD like Metallo Organic CVD (MOCVD), Atomic Layer Epitaxy (ALE), Vapour Phase Epitaxy (VPE), Plasma Enhanced CVD (PECVD). They differ in source gas pressure, geometrical layout and temperature used. Basic CVD process, however, can be considered as a transport of reactant vapour or reactant gas towards the substrate (see Fig. 3.16) kept at some high temperature where the reactant cracks into different products which diffuse on the surface, undergo some chemical reaction at appropriate site, nucleate and grow to form the desired material film. The by-products created on the substrate have to be transported back to the gaseous phase removing them from substrate.

Vapours of desired material may be often pumped into reaction chamber using some carrier gas. In some cases the reactions may occur through aerosol formation in gas phase. There are various processes such as reduction of gas, chemical reaction between different source gases, oxidation or some disproportionate reaction by which CVD can proceed. However it is preferable that the reaction occurs at the substrate rather than in the gas phase. Usually ~300 to 1200 °C temperature is used at the substrate. There are two ways viz. ‘hot wall’ and ‘cold wall’ by which substrates are heated as shown in Fig. 3.17.

In hot wall set up the deposition can take place even on reactor walls. This is avoided in cold wall design. Besides this, the reactions can take place in gas phase

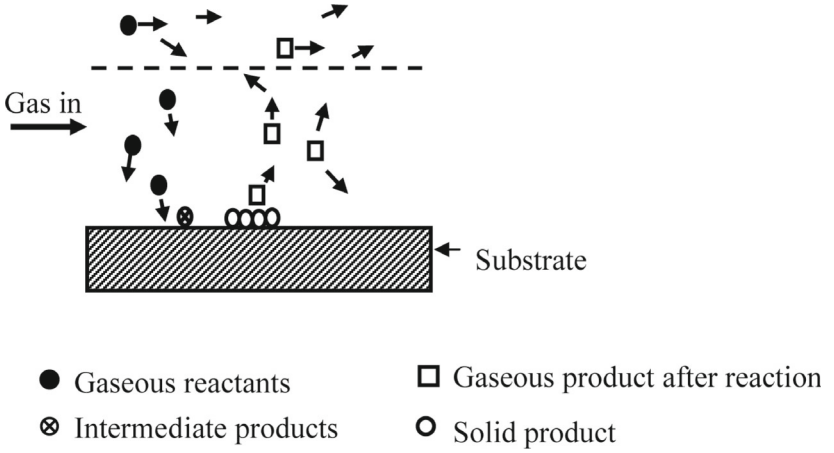


Fig. 3.16: Basic concept of Chemical Vapour Deposition (CVD) process.

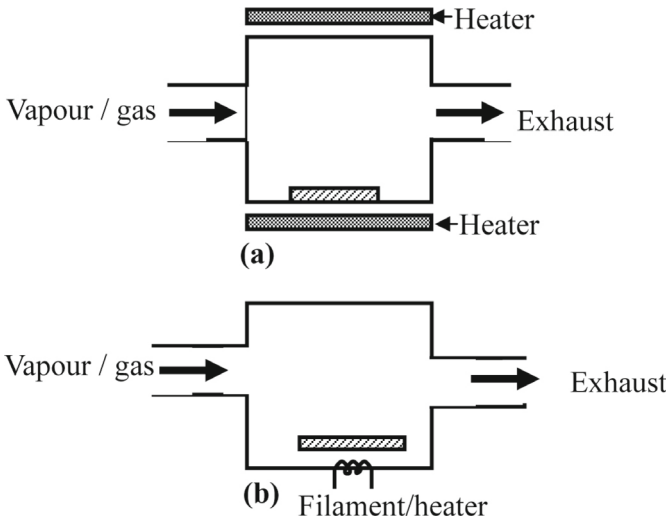


Fig. 3.17: (a) Hot wall and (b) cold wall CVD.

with hot wall design which is suppressed in cold wall set up. Further, coupling of plasma with chemical reaction in cold wall set up is feasible.

Usually gas pressures in the range of 100 Pa to 10^5 Pa are used. Growth rate and film quality depend upon the gas pressure and the substrate temperature. When the growth takes place at low temperature, it is limited by the kinetics of surface reaction. At intermediate temperature it is limited by mass transport i.e. supply of reacting gases to the substrate. Here the reaction is faster and supply of reactants is slower.

At high temperature, growth rate reduces due to desorption of precursors from the substrate.

When two types of atoms or molecules say *P* and *Q* are involved in the desired film, there are two ways in which growth can take place. In what is known as Langmuir–Hinshelwood mechanism, both *P* and *Q* type of atoms/molecules are adsorbed on the substrate surface and interact there to produce the product *PQ*. When one species is adsorbed in excess of the other, the growth depends on the availability of adsorption sites for both *P* and *Q* as shown in a schematic diagram (Fig. 3.18).

However it is also possible to have another way in which reaction can occur i.e. one species say *P* adsorbs on the substrate and the species *Q* from gas phase interacts with *P*. Thus there is no sharing of sites. This type of mechanism is known as Elay–Riedel mechanism as shown in Fig. 3.19.

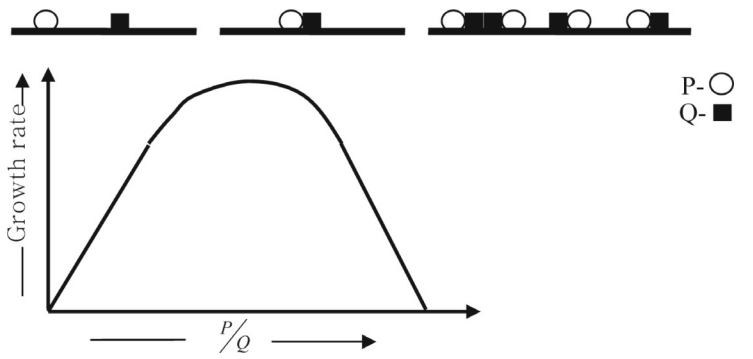


Fig. 3.18: Langmuir–Hinshelwood mechanism of growth.

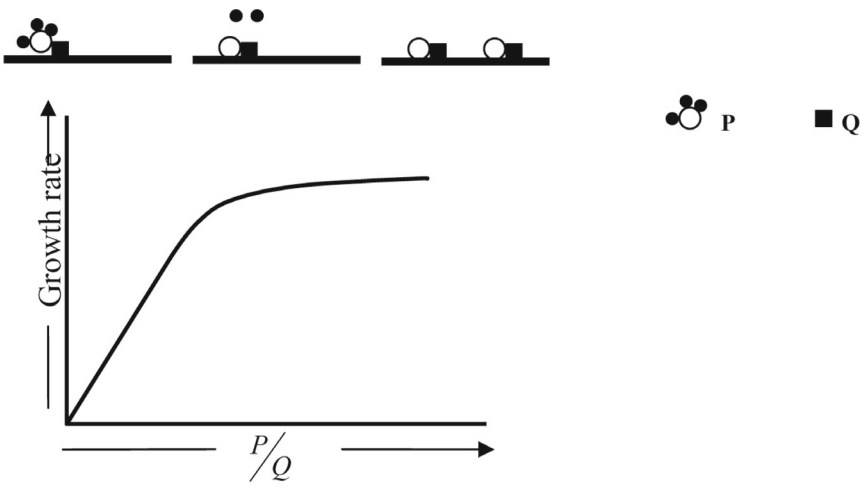


Fig. 3.19: Elay–Riedel mechanism.

3.6 ATOMIC LAYER DEPOSITION

Atomic Layer Deposition (ALD) is a thin film deposition technique as the name suggests to deposit atom thick layers. Although first conceptualized and published by Aleskovaskii and Kolt'sov way back in 1960, it is taking the momentum after the interest in nanotechnology was developed. The technique is essentially similar to Chemical Vapor Deposition (CVD) discussed in Sect. 3.5 in the sense that chemical vapors are allowed to interact with each other in a reactor (or furnace) maintained at some temperature and deposition on suitable substrate takes place. However unlike in CVD the precursor molecular gases are introduced in the reactor sequentially. When the first reacting molecular gas is introduced it is allowed to dissociate on the substrate. The excess first precursor and byproducts in gas phase are completely pumped out of the reactor. Then the second precursor is introduced which consequently interacts with the already deposited molecular fragments on the substrate. Once the second precursor has reacted completely, excess second precursor and byproducts in the gas phase are pumped out. If required there can be more than two chemical precursors to obtain reaction products on the substrate. This is first cycle. One can repeat the cycle of introducing the precursors to more than one cycle until the required film thickness is achieved. The sequence of process is schematically illustrated in Fig. 3.20.

In the schematic diagram, a substrate is shown as a rectangular piece. However, 0-D,1-D, 2-D, 3-D or conformal (any shape), porous substrates can be deposited as atomic layers. For uniform deposition sometimes the substrates are rotated around the axes or brought in sight of the inlet of precursors.

ALD is thus conceptually a simple process, in which precursors are sequentially pumped in the reactor and the reactor does not see any two or more precursors simultaneously being present in the reactor. The precursors, their introduction and

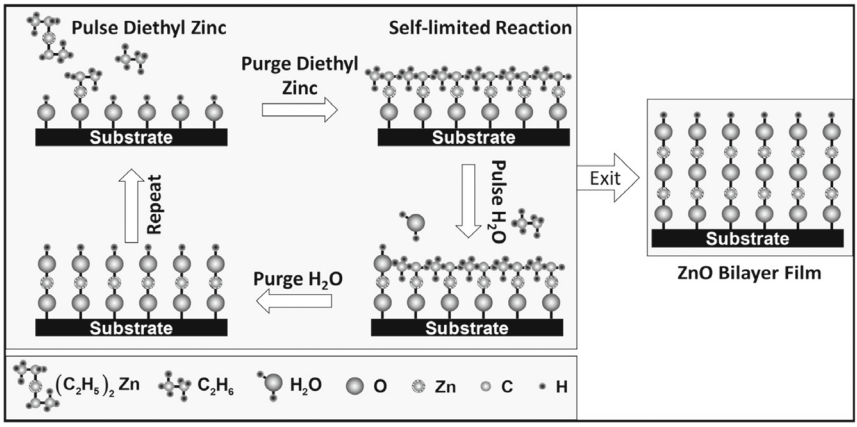


Fig. 3.20: Example of few layers of ZnO on a plane substrate by the Atomic Layer Deposition (ALD) technique. First precursor is H_2O and second precursor is $(C_2H_5)_2$ Zn.

removing timings need to be chosen properly. The precursors should not react with the reactor chamber or any inlet–outlet pipes. The intermediate products should be properly exhausted and disposed off as there can be toxic gases and nanoparticles in the gases being removed from the reactor.

The sequential deposition and removing the excess precursor and byproducts before another precursor comes in the reactor is a slow process. Therefore, compared to other methods of thin film growth, ALD is very slow. However, ALD provides a very fine atomic, sub-monolayer thickness and interface control as well as adhesion. It is possible with ALD to maintain requisite stoichiometry necessary for many processes like catalysts, sensors and medical field. The thin films obtained using ALD also provide defect and pinholes free thin films, which are useful in producing excellent, reproducible high electronic devices. However most of the drawbacks of the technique are related to the slow thin film growth and precursors have to be removed cyclically, slow process and controlled gas inlet–outlet requirements.

Yet the list of materials being deposited by ALD is growing now with the requirement of 2-D materials. So far some oxides and sulfides seem to have been grown on flat and 3-D substrates like carbon nanotube sponges, porous alumina, prous biological templates and so on. With the requirement of high precision of nano size memory devices ALD will play a decisive role.

3.7 ELECTRIC ARC DEPOSITION

This is one of the simplest and useful methods which leads to mass scale production of Fullerenes, carbon nanotubes (to be discussed in a later chapter) etc.

The set up as shown in Fig. 3.21 requires basically water cooled dome or vacuum chamber and two electrodes to strike an arc between them. The cathode itself acts as the source of material. If some catalysts are to be used, there can be some additional thermal source of evaporation. Depending upon the desired products inert gas or reactive gas introduction is necessary.

Usually the gap between the electrodes is ~1 mm and high current ~50 to 100 amperes is passed from a low voltage power supply (~12–15 V). Inert gas pressure is maintained in the vacuum system. When an arc is set up, cathode material evaporates. This is possible as long as the discharge can be maintained. The adjustment of the electrode gap without breaking the vacuum becomes essential, as one of the electrode burns and gap increases.

As will be discussed later, this method was the one in which by striking the arc between the two graphite electrodes, it was possible to get fullerenes in large quantity. Fullerenes are formed at low helium pressure and nanotubes are formed at high pressure. Also, fullerenes are obtained by purification of soot collected from inner walls of vacuum chamber, whereas nanotubes are found to be formed only at high He gas pressure and in the central portion of the cathode. No carbon nanotubes are found on the chamber walls. Some nanoparticles of carbon also are usually found around the region where nanotubes are formed. Temperature reaches as high

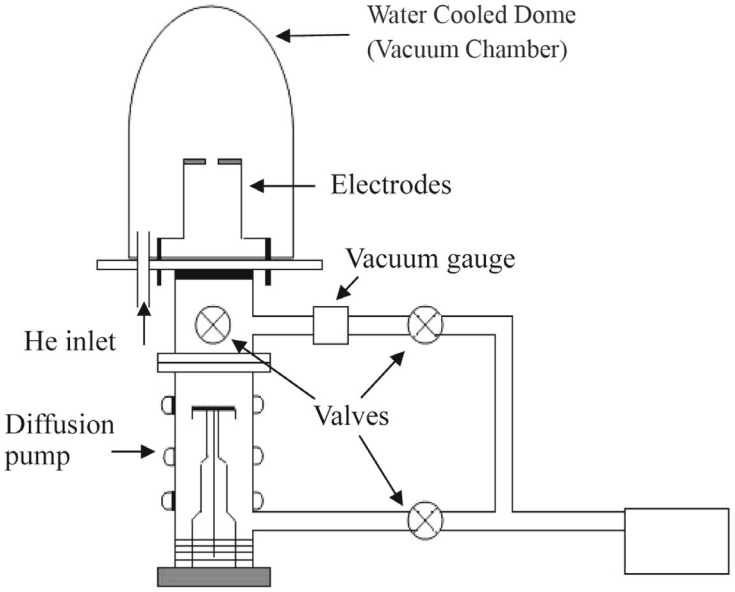


Fig. 3.21: Arc deposition set up.

as ~3500 °C as arc discharge takes place. In principle, other nanocrystals or tubes also should be possible to be obtained by this method. However this method is mostly found to be suitable for deposition of fullerenes or carbon nanotubes.

3.8 ION BEAM TECHNIQUES (ION IMPLANTATION)

There are many examples in which high energy (few keV to hundreds of keV) or low energy (<200 eV) ions are used to obtain nanoparticles. Ions of interest are usually formed using an ion gun specially designed to produce metal ions which are accelerated with high or low energy towards the substrate heated to few hundreds of ^{o}C. Depending upon the energy of the incident ions, various other processes like sputtering and electromagnetic radiation may take place as already discussed in connection with sputter deposition in Sect. 3.3. It is possible to obtain single element nanoparticles or compounds and alloys of more than one element. Post-annealing also is used sometimes to improve the crystallinity of the materials. In some experiments it has been possible to even obtain doped nanoparticles (i.e. nanoparticles in which some foreign atoms are intentionally introduced to alter the properties of the host material in the controlled fashion) using ion implantation.

There is also a possibility of making nanoparticles using swift heavy ions (few MeV energy) employing high energy ion accelerators like pelletrons.

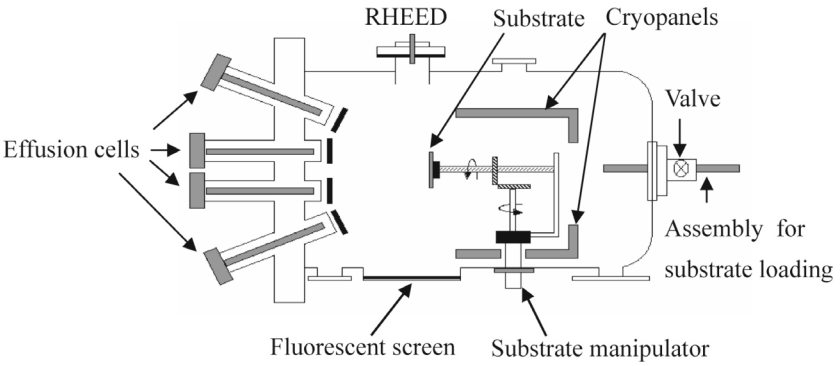


Fig. 3.22: Schematic diagram of molecular beam epitaxy.

3.9 MOLECULAR BEAM EPITAXY (MBE)

This technique can be used to deposit elemental or compound quantum dots, quantum wells as well as quantum wires in a very controlled manner. A schematic diagram is given in Fig. 3.22. High degree of purity is achievable using ultra high vacuum (better than 10^{-8} Pa). Special sources of deposition known as Knudsen cell (K-cell) or effusion cell are employed to obtain molecular beams of the constituent elements. The rate of deposition is kept very low and substrate temperature is rather high in order to achieve sufficient mobility of the elements on the substrate and layer by layer growth to obtain nanostructures or high purity thin films. Technique like Reflected High Energy Electron Diffraction (RHEED) is incorporated to monitor the high crystallinity of the growing film.

FURTHER READINGS

Peter Ozaveshe Oviroh, Rokhsareh Akbarzadeh, Dongqing Pan, Rigardt Alfred Maarten Coetzee and Tien-Chien Jen. *Sci and Tech. of Adv. Mater*., **20:** 465 (2019).

Material Science of Thin Films, Deposition and Structure by M. Ohring, Second Edition. Academic Press, Boston (USA) (2002).

Cluster Beam Synthesis of Nanostructured Materials by P. Milani and S. Iannotta. Springer Verlag, Berlin, Heidelberg, New York (1999).

R. Vyeda. *Prog. Mater. Sci*., **35:** 1 (1991).

H. Gleiter. *Prog. Mater. Sci*., **33:** 223 (1989).

Handbook of Thin Film Technology by L.I. Maissel and R. Glang. McGraw-Hill Book (USA) (1970).

Thin Films Phenomenon by K.L. Chopra. McGraw-Hill Book (USA) (1969).

Chapter 4
Synthesis of Nanomaterials—II (Chemical Methods)

4.1 INTRODUCTION

In this chapter we shall discuss some of the wet chemical methods to synthesize nanomaterials. There are numerous advantages of using chemical methods which are summarized in Box 4.1. In some cases nanomaterials are obtained as colloidal particles in solutions, which can be filtered and dried to obtain powder. We can obtain thin films or nanoporous materials by electrodeposition and etching. Advantages of chemical synthesis are manifold. In many cases very well known chemical reaction route can be optimized to obtain nanoparticles. Particles of different shapes and sizes are possible depending upon the chemicals used and reaction conditions.

Box 4.1
Some Advantages of Chemical Synthesis

- Simple techniques
- Inexpensive, less instrumentation compared to many physical methods
- Low temperature (<350 °C) synthesis
- Doping of foreign atoms (ions) possible during synthesis
- Large quantities of the materials can be obtained
- Variety of sizes and shapes are possible
- Materials are obtained in the form of liquid but can be easily converted into dry powder or thin films
- Self assembly or patterning is possible

Low to high temperature routes are possible. In many cases, particles can be doped with different metal ions quite easily. Coupled, coated, chemically capped (by some molecules or passivated) particles can be made. Most important is that very narrow size distributed materials are possible to synthesize and in many cases large quantities of materials can be obtained. The instrumentation involved in the chemical synthesis

can be relatively simple and inexpensive as compared to many physical methods. As in many cases nanoparticles synthesized by chemical method form what is known as 'colloids'. We shall first try to understand them and then proceed to some specific chemical routes to obtain them as nanoparticles.

4.2 COLLOIDS AND COLLOIDS IN SOLUTIONS

Colloids are known since very long time. A class of materials, in which two or more phases (solid, liquid or gas) of same or different materials co-exist with the dimensions of at least one of the phases less than a micrometre is known as colloids. Colloids may be particles, plates or fibers (see Fig. 4.1). Nanomaterials are a sub-class of colloids, in which one of the dimensions of colloids is in nanometer (~1 to 100 nm) range.

There are several examples around us, having different combinations of phases, in the form of colloids like liquid in gas (fog), liquid in liquid (fat droplets in milk), solid in liquid (tooth paste), solid in solid (tinted glass), gas in liquid (foam). There can be multiple existing colloids like water and oil bubbles in porous mineral rocks. Organic and inorganic materials can be dispersed into each other to form colloids. Several examples exist even of bio-colloids. Blood and bones are good examples of bio-colloids. Blood has corpuscles dispersed in serum and bone has colloids of calcium phosphate embedded in collagen.

Colloids may even form networks. For example aerogels (discussed in more details in Chapter 9) are a network of silica colloidal particles, pores of which are filled with air.

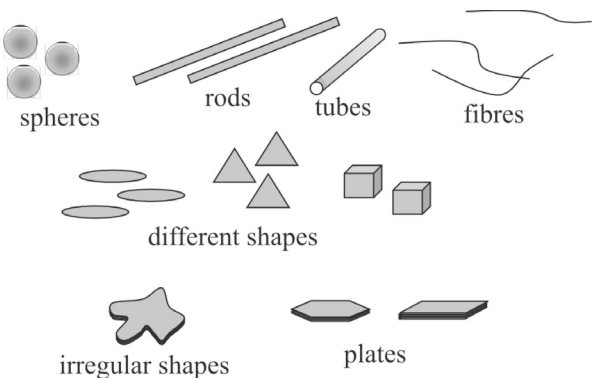


Fig. 4.1: Different shapes of colloids.

4.2.1 Interactions of Colloids

Colloids are particles with large surface to volume ratio. Correspondingly there are large number of atoms/molecules on the surface of a colloidal particle, which do not have as many neighbours as those for an atom/molecule inside the interior. Therefore atoms on the surface are in a highly reactive state, which easily interact to form bigger particles or tend to coagulate. It is thus necessary to understand the stability of colloids i.e. how the colloids dispersed in a medium can remain as separated particles. In general there are a number of interactions involved. For the sake of understanding these interactions, we consider the inorganic spherical colloids of equal size, dispersed in a liquid medium. When fine particles are dispersed in a liquid medium, it is known that they undergo *Brownian motion.* If we are able to tag a particle in the solution, as depicted in Fig. 4.2, it would appear as if it is making a random motion. All other particles also execute random motion, hitting each other and changing direction of motion in solution. Distance travelled between successive collisions is random too. However an average distance travelled by a colloidal particle can be found as

$$\Delta\overline{R}^2 = \left(\frac{kT}{3\pi r\eta}\right)\Delta t \tag{4.1}$$

where $\Delta\overline{R}$ is distance travelled by a particle from its original position in time Δt, k – Boltzmann's constant, T – temperature of liquid, r – particle radius and η is viscosity of the liquid.

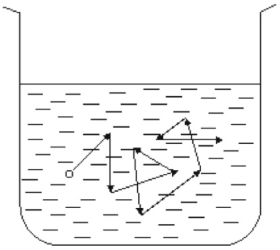


Fig. 4.2: Brownian motion of colloidal particles.

Box 4.2
Robert Brown (1773–1858)

Robert Brown who discovered Brownian motion in 1827 was born on 21st December 1773 in Montrose, Scotland. He studied medicine at the University of Edinburgh. Immediately after completing his education he joined Fifeshire Regiment of Fencibles as Ensign and Surgeon's Mate. In his free time he used to study botany. On 18th July 1801 he joined an eminent botanist Sir Joseph Banks to go to Australia on an

expedition. Their goal was to collect some rare plants and study them. They returned back to England in October 1805 with a huge collection of some 4000 species of plants, some zoological specimens and numerous drawings and notes. Brown spent next five years working on this material. Brown used a microscope throughout his studies. He is considered to be a 'gifted observer'. He identified naked ovule in the gymnosperm which is a rather difficult task even using modern microscopes. He however is famous due to 'Brownian motion' which he observed in pollen grains using his simple optical microscope. Others also had observed such motion of particles under microscope but they simply attributed it to 'life' itself. Careful experiments by Brown showed that they were not the consequence of 'life' nor any currents in fluids or evaporation of liquid.



From 1806 to 1822 Robert Brown served as a clerk, librarian and housekeeper of the Linnen Society of London. He was elected as fellow of Royal Society in 1810 and fellow of Linnen Society in 1822. He remained the president of Linnen Society from 1849 to 1853. He died on 10th June 1858 in London.

Interactions of such constantly and randomly moving particles with each other and with liquid in general would be quite complex. Consider for example the van der Waals interaction (see Chapter 2) between two atoms or molecules. There are two parts in the interaction viz. attractive and repulsive parts given by the equation (4.2), irrespective of whether there exist permanent dipoles or not. The interaction is given as

$$dG_1 = \left(\frac{A}{R^{12}}\right) - \left(\frac{B}{R^6}\right) \tag{4.2}$$

where dG_1 is the interaction energy, A and B are constants and R is the distance between two particles.

Here the first term is repulsive interaction (Born repulsive interaction) effective only at short distance and second term represents long range attractive interaction (van der Waals attraction). Repulsive part arises due to repulsion between electron clouds in each atom and attractive part is due to interaction between fluctuating

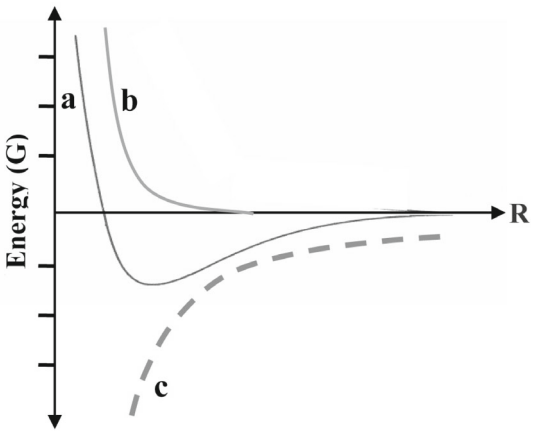


Fig. 4.3: Van der Waals interaction: (a) Resulting energy, (b) Repulsive energy and (c) Attractive energy.

or permanent dipoles of atoms/molecules. Schematically it is shown in Fig. 4.3. Equation (4.2) is known as Lennard-Jones equation.

Box 4.3
Free Energy

Free energy of a body is a measure of its ability to do work. A body always tends to attain the state of lower energy by releasing 'free energy' and is given by Gibb's free energy

$$G = H - TS \tag{4.3}$$

where G is the energy absorbed or released by a body, H – enthalpy, T – temperature and S is entropy.

In order to understand the meaning of surface free energy, consider that a cylinder with cross section 'A' and length '$2l$' is cut into two equal pieces as shown in Fig. 4.4. Work has to be done in order to break the cylinder into two pieces and separate them by a distance R.

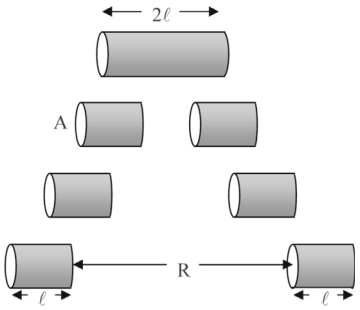


Fig. 4.4: A cylinder with cross section A and length 2*l* has been cut into two pieces of equal length.

Energy increases as the distance between the two pieces increases. The two pieces keep on attracting each other due to intermolecular forces at the broken interface. Therefore, more and more energy has to be supplied in order to separate them. Interfacial energy is related to the surface tension 'γ'. Surface tension γ is along the normal to surface and is expressed as force per unit length (N/m or Joule/m^2). Surface free energy is given by

$$\Delta W = \Delta G - 2\gamma A \tag{4.4}$$

where ΔW is the work done to break the cylinder, γ–surface tension and A is the surface area. (Factor '2' arises because while creating two surfaces, surface area would be 2A for two surfaces as shown in Fig. 4.4.)

The surface free energy per unit area would increase as shown in Fig. 4.5 with separation R between two particles.

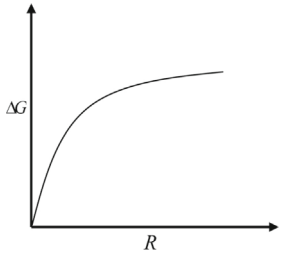


Fig. 4.5: Change in surface free energy with separation R between two pieces of the cylinder shown in Fig. 4.4.

The force of attraction would increase in vacuum as shown in Fig. 4.6. In a liquid the force of attraction would reduce in general.

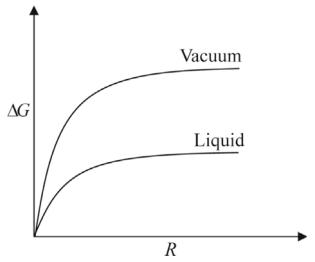


Fig. 4.6: Change in surface free energy with separation R between two pieces of the cylinder shown in Fig. 4.4, placed in vacuum or liquid.

4.2.2 Colloids in Vacuum

Lennard-Jones equation (4.2) is sufficient to describe an interaction between two atoms or molecules. When we consider colloids with large number of atoms, we need to take into account all the atoms and their interactions with each other. This is quite a complex situation as shown in Fig. 4.7. To describe the interaction between colloidal particles Deraguin, Landau, Verwey and Overbeck proposed a theory known as DLVO theory. In order to reduce the complexity of the problem, they assumed two interacting spherical particles of equal size. Let the radius of each particle be 'r' and let two particles be separated by a distance 'R'.

It was shown that for two similar spherical particles in vacuum the attractive interaction is given by

$$dG \text{ (attraction)} = -\left(\frac{A_H \cdot r}{12R}\right)\left[1 + \frac{3}{4} \cdot \frac{R}{r}\right] \tag{4.5}$$

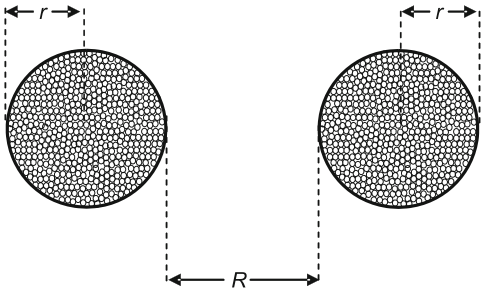


Fig. 4.7: Interaction even between two spherical particles of same material and same size is complex due to presence of large number of atoms in each particle.

where A_H is known as Hamaker constant and is given by

$$A_{\mathrm{H}} = A'\pi^2 n^2 \tag{4.6}$$

where A' is a constant related to A in the Lennard-Jones equation (4.2) and n is the number of atoms/molecules per unit volume in a colloid.

4.2.3 Colloids in a Solution

So far we considered the colloids just in vacuum. Consider now a situation, in which inorganic spherical colloids are immersed in a liquid (and do not dissolve). The attractive interactions between the colloids get modified through the change of Hamaker constant as A_{H}, which can be written now as

$$A_{\mathrm{H}} = \left(\sqrt{A_{1v}} - \sqrt{A_{2v}}\right)^2 \tag{4.7}$$

where A_{1v} is the Hamaker constant for particle of inorganic solid under consideration, in vacuum and A_{2v} is Hamaker constant of colloid of medium in vacuum. It can be seen from above equation that in general the effect of liquid medium is to reduce the Hamaker constant of colloid particle. Hence the attractive force between colloid particles will in general reduce.

4.2.4 Effect of Charges on Colloids

Colloids in liquid may be positively charged, negatively charged or even neutral. But in most of the cases they are charged. There are various sources for colloids by which they acquire charges on their surfaces viz. through composition of colloidal material, properties of dispersing medium including the type and concentration of dissolved ions in the solution. In any case as soon as there are some charges on particles, ions of opposite charges accumulate around them. Oppositely charged ions are known as *counter ions.*

This accumulation of counter ions leads to formation of an electric double layer. Helmoltz considered that the situation is like that in a parallel plate condenser, where there are opposite charges as plates separated by a medium (for example air as shown in Fig. 4.8). Due to their Brownian motion, counter ions are not fixed nor are colloidal particles at rest. They execute their own Brownian motion and form a dynamic double layer around them as schematically shown in Fig. 4.9.

The changes occur when concentrations of charges on colloid and local ionic charges in solution contribute to double layer change. The concentration of electrolyte

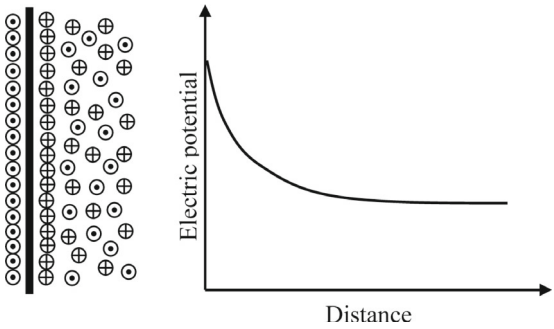


Fig. 4.8: Double layer of charges and fall of electrical potential from negatively charged plate with positive ion accumulation.

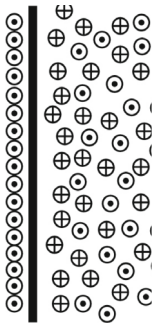


Fig. 4.9: Diffuse double layer because of Brownian motion.

would strongly affect the electric potential curve and is shown schematically in Fig. 4.10.

Consider now a situation in which two charged colloidal particles come closer with their electric double charge layers. As they approach each other force of repulsion increases as shown in Fig. 4.11. It is easy to see that the difference in concentrations of double layer charges would play an important role.

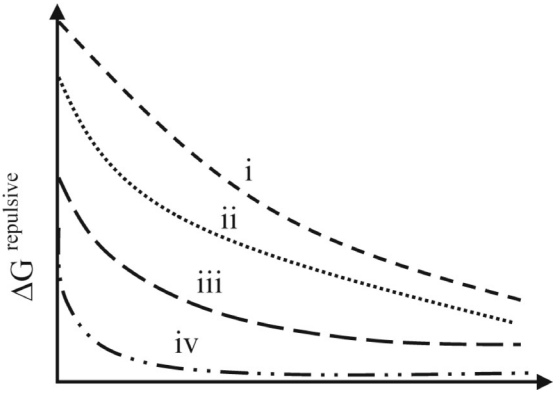


Fig. 4.10: Effect of concentration on change of potential from a particle. Curves i, ii, iii and iv refer to increasing electrolyte concentrations.

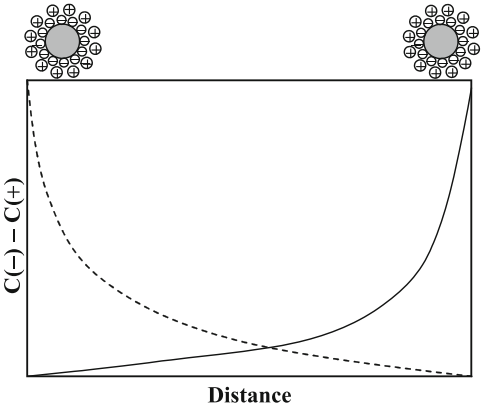


Fig. 4.11: Colloid-colloid interaction with electric double charge layer.

The repulsive interaction for low concentration would set at longer distance and that for higher concentration at smaller distance.

4.2.5 Stearic Repulsion

Stability of colloids can be increased by creating Stearic hindrance or repulsion (see Fig. 4.12).

By adsorbing some layers of a different material on colloidal particles e.g. polymer or organic molecules on inorganic colloidal particles, it is possible to reduce the attractive forces between them. With addition of adsorbed layers, the effective sizes of the particles change which helps them to stay at a longer distance from each other, reducing the attractive interaction. However, in case where the coating material is similar in properties to that of the solvent, the effect of coating would be negligible. By anchoring long chain molecules on the particles, it is possible to keep them apart with negligible interaction. This idea is the basis of 'capped nanoparticles' discussed in the next section.

Thus the colloids interact with each other dynamically and are affected by van der Waals forces, colloid-colloid interaction mediated through dispersing medium,

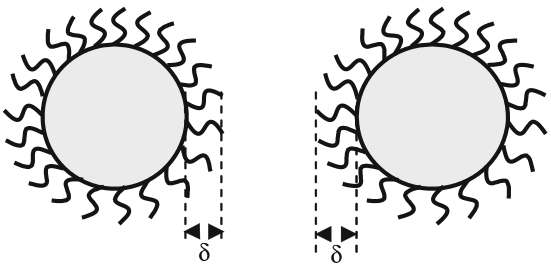


Fig. 4.12: Colloids with coatings. Separation between the particles, each coated with coating of thickness δ, increases by 2δ.

electric double layer and Stearic interactions. All the interactions may not be set in for every case. In general, contributions of these various attractive and repulsive interactions—also dictated by temperature, concentration of colloids and dispersing medium—are additive and can be written as follows.

$$\begin{aligned}\Delta G = {} & \Delta G_1 \text{ (attractive and repulsive)} \\ & + \Delta G_2 \text{ (colloid-colloid attractive)} \\ & + \Delta G_3 \text{ (electrostatic repulsion)} + \Delta G_4 \text{ (Stearic repulsion)} \\ & + \Delta G_5 \text{ (any other)}\end{aligned} \tag{4.8}$$

If the repulsive forces are strong enough, colloids would be stabilized. Otherwise ripening, coagulation, or network formation may take place.

4.3 CHEMICAL SYNTHESIS OF NANOPARTICLES

Colloids are thus phase separated submicrometre particles in the form of spherical particles or particles of various shapes and sizes like rods, tubes, plates. They are the particles suspended in some host matrix. Metal, alloy, semiconductor and insulator particles of different shapes and sizes can be synthesized in aqueous or non-aqueous media. Colloidal particles in liquids are stabilized as discussed above by Coulombic repulsion, which arises due to similar charges they may have acquired on their surfaces. In some cases surface passivating molecules may be used which provide sufficient stearic hindrance inhibiting coalescence or aggregation. Nanomaterials are a special class of colloidal particles that are one to hundred nanometer size.

Synthesis of nanoparticles by chemical route is a very old method. Making nanoparticles using colloidal route goes back to nineteenth century when M. Faraday synthesized (in 1847) gold nanoparticles by wet chemical route. The particles are so stable that even today the colloidal solution made by him can be seen in the British Museum in UK. Unfortunately due to lack of appropriate microscopes at that time to resolve the tiny nanoparticles, he could not explain that the observed magenta colour of the gold solution was due to gold nanoparticles.

Here we shall discuss some commonly used synthesis methods of metal, semiconductor and insulator nanoparticles with some examples.

Chemical reactions are carried out in some glass vessel to synthesize nanoparticles of suitable size (less than ~ 100 nm). Glass vessel usually has a provision to introduce some precursors, gases as well as measure temperature and pH during the reaction. It is usually possible to remove the products at suitable time intervals. Reaction is usually carried out under inert atmosphere like argon or nitrogen gas so as to avoid any uncontrolled oxidation of the products. There is also a provision made to stir the reactants during the reaction by using teflon-coated magnetic needle. Figure 4.13

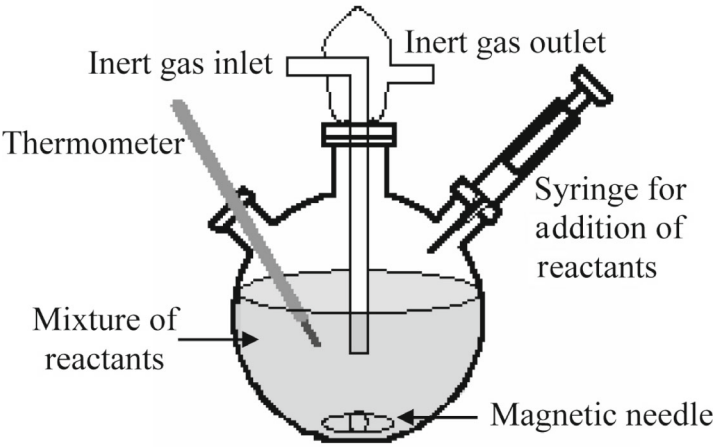


Fig. 4.13: A typical chemical reactor to synthesize nanoparticles.

illustrates a simple chemical synthesis set up to obtain nanoparticles by colloidal route.

To obtain high quality, crystalline, monodispersed (particles of same size), reproducible nanoparticles is often a very difficult task due to very sensitive nucleation and growth process discussed below. In the nanoparticles synthesis obtaining particles of uniform size is very crucial as their properties are size dependent. Large deviation from one size would mean change in the observed property. Therefore methods of 'size selective precipitation (see box 4.4)' have to be adopted.

4.3.1 Nucleation and Growth of Nanoparticles

Synthesis of nanoparticles of different shapes and sizes may appear as a complex process. Over several decades, scientists have tried to understand the process of atom-by-atom nucleation and growth of small to large particles in melts, aqueous or non-aqueous media from gas phase, or even in solids. The process of nucleation is a 'bottom-up' approach in which atoms and/or molecules come together to form a solid. The process can be spontaneous, and it may be homogenous or heterogeneous nucleation. Homogenous nucleation is said to take place when it involves the nucleation around the constituent atoms or molecules of the resultant particles. Heterogeneous nucleation, on the other hand, can take place on a foreign particle like dust, deliberately adding seed particles, templates, or the walls of the container.

Nucleation can also occur by a cavitation process. In cavitation, if there are some bubbles (which can be formed deliberately, as discussed in a later section) in the solution which then collapse, the high local temperature and pressure thus generated may be sufficient to cause homogenous nucleation. It can be seen in curve A of

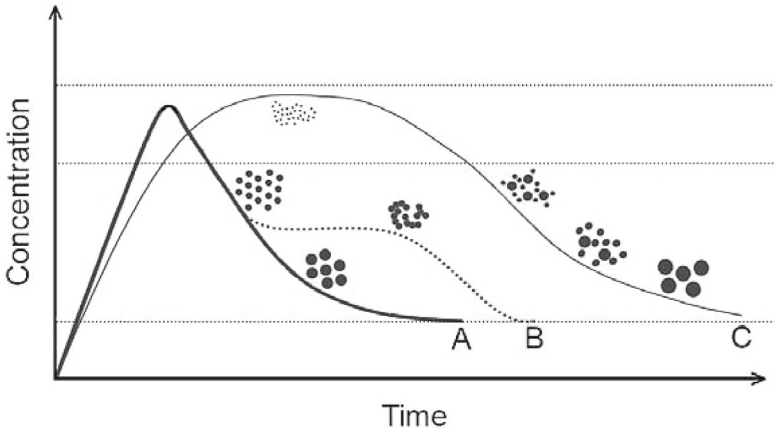


Fig. 4.14: Nucleation and growth of nanoparticles (LaMer diagram). All the dots appearing in the sketch are small/big nanoparticles.

Fig. 4.14 that fast nucleation takes place as the solute concentration approaches super saturation. If the nuclei acquire atoms quickly by diffusion through solution thus reducing the solute concentration, particles of uniform size are formed in a relatively shorter time compared to aggregated particles in curve B, or *Ostwald-ripened* particles as in C.

In the Ostwald ripening process, the nucleation proceeds for a long time bringing in smaller and larger nuclei to co-exist when super-saturation region exists over a longer period and then the solute concentration decreases. Larger particles tend to grow at the expense of smaller ones, thus becoming even larger. This is because of the lowering of total surface energy. It should be remembered that the concentration of solutes and temperature of the solution would strongly affect the growth. Additionally, the crystalline structure, defects, favourable sites, etc. would strongly affect the final products.

As schematically illustrated in Fig. 4.14, once the nuclei are formed they may take different routes—A, B, or C—depending on the growth conditions. The growth route depicted by curve A is a classical route suggested by LaMer, and is hence called as a LaMer diagram.

Nucleation process is controlled (except in bio-minerals, to be discussed in Chapter 5) thermodynamically. The size of a nucleus is determined by both the free energy change occurring during the formation of the solid (from a liquid), as well as the surface energy of the nucleus. A stable nucleus (of a critical radius r^*) needs to be formed so that it can grow into a larger stable particle. Let us refer to particles of radii smaller than r^* as embryos; the energy for such an embryo formation (ΔG_r) is given by:

$$\Delta G_r = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma_{SL} \tag{4.9}$$

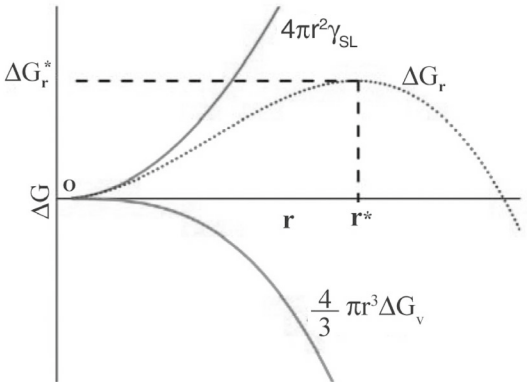


Fig. 4.15: Nucleation process.

where r is the radius of the embryo, ΔG_v is the free energy change per unit volume between the liquid and solid, and γ_{SL} is the interfacial free energy of the liquid and solid.

Below the melting temperature (T_m) of the solid, ΔG_v is negative, whereas the surface free energy, or surface tension γ_{SL} is positive. The two energies compete with each other with increasing value of the embryo radius r. The nature of the resulting curve for ΔG_r is illustrated schematically in Fig. 4.15.

Differentiating equation (4.9) with respect to r and equating it to zero at $r = r^*$, we obtain:

$$4\pi r^{*2}\Delta G_v + 4\pi\left(2r^*\right)\gamma_{SL} = 0 \tag{4.10}$$

Hence,

$$r^* = \frac{-2\gamma_{SL}}{\Delta G_v} \tag{4.11}$$

After the critical size r^*, the free energy starts decreasing and the growth begins.

The energy ΔG_v depends on latent heat of fusion and the degree of undercooling. Undercooling is a result of faster cooling rate than required for the equilibrium cooling. Some finite time is required in any system so that atoms/molecules adjust themselves and acquire the position of minimum energy. The undercooling temperature is given as:

$$\Delta T = T - T_m \tag{4.12}$$

where T_m is equilibrium melting temperature, T – bulk temperature, and ΔH_f is the heat of fusion per unit volume.

It can be shown that

$$\Delta G_v = \frac{\Delta H_f \Delta T}{T_m} \tag{4.13}$$

Therefore,

$$r^* = \frac{-2\gamma_{SL} T_m}{\Delta H_f \Delta T} \tag{4.14}$$

If ΔG_v is the total free energy of the nucleus of radius r, number of clusters n_r can be obtained using

$$n_r = Nexp\left(\frac{-\Delta G_v}{kT}\right) \tag{4.15}$$

where N is the total number of atoms and k is Boltzmann constant.

In case there is strain caused in the nucleus formation, additional term $\frac{4}{3}\pi r^3 \varepsilon$ can be added to the equation (4.9). Here ε is the strain energy and is positive.

When the nucleation occurs on some foreign particle or surface (e.g. of the container wall or substrate), heterogeneous nucleation is said to occur. This lowers the energy necessary for the nucleation of a particle. Consequently the critical size or radius r^* for nucleation is smaller in heterogeneous nucleation than that for homogeneous nucleation. This can be understood as follows.

Consider a solid substrate on which a drop of liquid forms as illustrated in Fig. 4.16. γ_{Sd}, γ_{SL} and γ_{dL} are the interfacial surface tensions between substrate and liquid drop, surface and liquid and the drop and liquid respectively. It can be seen from Fig. 4.16 that:

$$\gamma_{dL} \cos\theta = \gamma_{SL} - \gamma_{Sd} \tag{4.16}$$

Therefore,

$$\Delta G = V \Delta G_v + A_{dL}\gamma_{dL} - \pi r^2 \gamma_{dL} \cos\theta \tag{4.17}$$

where V is the volume of the liquid drop and A_{dL} is interfacial surface area between the drop the liquid; it may be noted that there is change in the volume free energy ΔG_v

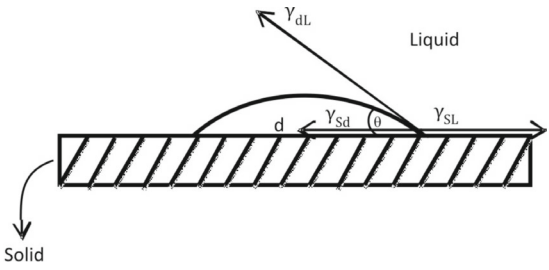


Fig. 4.16: Nucleation on a solid surface.

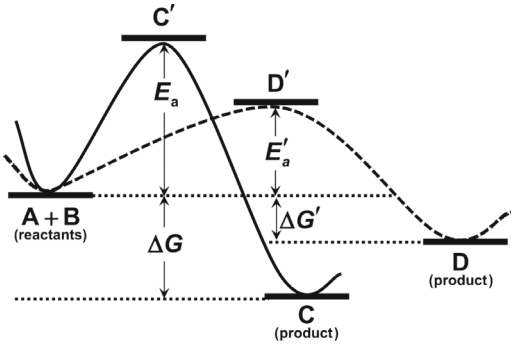


Fig. 4.17: Thermodynamic (solid curve) and kinetic pathway (dashed like curve). Ea and Ea' are activation energies. Free energies ΔG and ΔG' also are shown on the thermodynamic and kinetic path curves.

as discussed with reference to homogeneous nucleation. Hence the critical radius r^* which is given as

$$r^* = \frac{-2\gamma_{dL}}{\Delta G_v} \tag{4.18}$$

also changes. In general, once the nuclei with critical radii r^* are generated, stable nuclei and particle growth starts by addition of atoms or molecules from the solute.

In the nanoparticle synthesis the control may or may not be thermodynamic but kinetic also. In the kinetic control the reaction pathways take place to stabilize the reaction product in a lower stable state compared to thermodynamically stable state as schematically illustrated in Fig. 4.17. Various factors such as surfactants, concentration of intentional and unintentional additives, pH, temperature and so on are responsible for alternate paths rather than thermodynamically most stable state path. It is usually very difficult to simultaneously regulate number of kinetic controls resulting into non-reproducibility of synthesis recipes. Understanding of the kinetic controlling factors is very important. Figure 4.14 shows different pathways during the growth of nanoparticles. Although very interesting, further discussion is beyond the scope of this book. However, in Chapter 15 (Practicals) examples of anisotropic nanoparticles produced with kinetic controls are given.

4.3.2 Synthesis of Metal Nanoparticles by Colloidal Route

Metal nanoparticles are often synthesized by reduction of some metal salt or acid. For example highly stable gold particles can be obtained by reducing chloroauric acid ($HAuCl_4$) with tri sodium citrate ($Na_3C_6H_5O_7$). The reaction takes place as follows (Fig. 4.18):

$$HAuCl_4 + Na_3C_6H_5O_7 \rightarrow Au^+ + C_6H_5O_7^- + HCl + 3NaCl \tag{4.19}$$

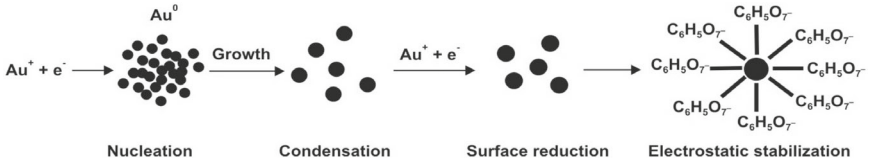


Fig. 4.18: Stabilization by electrochemical double layer formation.

The reaction can be carried out in water using the set up shown in Fig. 4.13.

Metal gold nanoparticles exhibit intense red, magenta and other colours, depending upon the particle size. The size dependent optical properties of metal nanoparticles are discussed in Chapter 8. Gold nanoparticles discussed above are stabilized by repulsive Coulombic interactions. It is also possible to stabilize gold nanoparticles using thiol or some other capping molecules.

In a similar manner, silver, palladium, copper and other metal nanoparticles can be synthesized using appropriate precursors, temperature, pH, duration of synthesis etc. Particle size, size distribution and shape strongly depend on the reaction parameters and can be controlled to achieve desired results. It is also possible to synthesize alloy nanoparticles using appropriate precursors.

4.3.3 Synthesis of Semiconductor Nanoparticles by Colloidal Route

Compound semiconductor nanoparticles can be synthesized by wet chemical route using appropriate salts. Here we give some examples to obtain semiconductor nanoparticles.

Sulphide semiconductors like CdS and ZnS can be synthesized as nano-particles simply by coprecipitation. For example, to obtain ZnS nanoparticles any zinc salt like zinc sulphate ($ZnSO_4$), zinc chloride ($ZnCl_2$), zinc nitrate ($ZnNO_3$) or zinc acetate ($Zn[CH_3COO]_2$) can be dissolved in aqueous medium and Na_2S is added to the solution. (One can even dissolve H_2S gas in the Zn salt solution.)

Following simple reaction

$$ZnCl_2 + H_2S \rightarrow ZnS + 2HCl \tag{4.20}$$

results to give solid colloidal particles of ZnS. To obtain zinc oxide particles one can use NaOH. Following reaction takes place.

$$ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$$

$$Zn(OH)_2 \rightarrow ZnO + H_2O \tag{4.21}$$

Box 4.4
Chemical Capping of Nanoparticles

A variety of molecules can be used to cap the nanoparticles. For example capping of metal-sulphide nanoparticles by few organic and inorganic molecules is illustrated in Fig. 4.19.

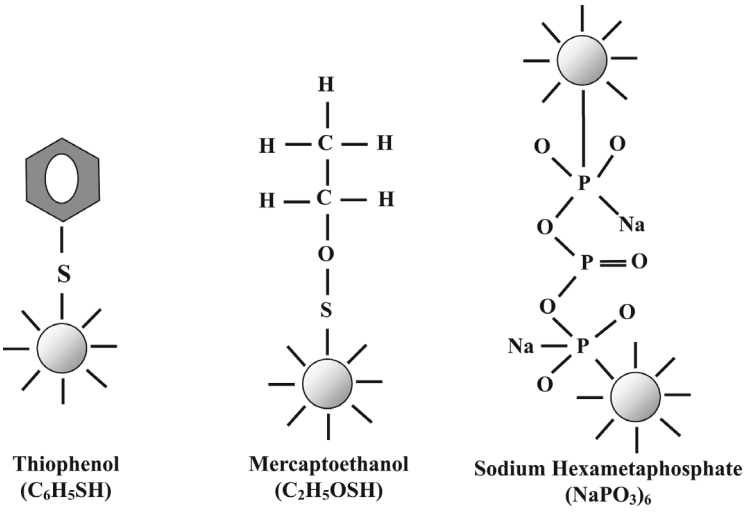


Fig. 4.19: Capping of nanoparticles by different molecules.

In another variation of chemical capping method, reactions of inexpensive and non-toxic chemicals like metal chlorides, nitrates, acetates and inorganic salts are performed at moderate temperatures like 80 to 120 °C. In some cases, it is even possible to synthesize nanoparticles at room temperature. However the initial size distribution can be quite broad. The size distribution can be narrowed down by a method known as 'size selective precipitation'. For this one needs to use a proper pair of solvent-nonsolvent liquids. Some of the solvent-nonsolvent pairs are pyridine-hexane, chloroform-methanol or dimethyl sulphoxide-diethyl ether.

The nanoparticles are dispersed in a solvent so as to get an optically clear solution. Nonsolvent solution is then added so that flocculation occurs. Supernatent and flocculate are separated by centrifugation. Precipitate has larger particles and can be separated. Bigger particles, therefore, are first separated from smaller particles and again dispersed in the solvent solution. The process is continued until no change in the size distribution is observed by repeating the procedure.

Advantage of chemical capping method is mainly that nanoparticles are chemically stable over a long time. The thermal stability depends upon the capping molecules used. In most cases, where organic molecules are used, particles are stable upto about 200-250 °C and may find considerable range of applications. Another advantage with both the methods is that while synthesizing the nanoparticles, they can be doped with

some metal ions so as to get fluorescent particles at relatively low temperatures. A wide range of semiconductor nanoparticles can be synthesized by this way and are found to be useful in many applications.

Selenide particles can be obtained using appropriate selenium giving salt.

However all these nanoparticles need to be surface passivated as colloids formed in liquids have a tendency to coagulate or ripen due to attractive forces existing between them. The electrostatic and other repulsive forces may not be sufficient to keep them apart. However, as it was also discussed earlier, stearic hindrance can be created by appropriately coating the particles to keep them apart. This is often known as 'chemical capping' and has become a widely used method in the synthesis of nanoparticles. Advantage with this chemical route is that, one can get stable particles of variety of materials not only in the solution, but even after drying off the liquid. One can even make thin films of the capped particles by spin coating or dip coating techniques. The coating, however, has to be stable and non-interactive with the particle itself except at the surface. Coatings may be a part of post-treatment or a part of the synthesis reaction to obtain nanoparticles. If it is a part of the synthesis reaction, the concentration of capping molecules can be used in two ways i.e. to control the size as well as to protect the particles from coagulation.

Chemical capping can be carried out at high or low temperature depending on the reactants. In high temperature reactions, cold organometallic reactants are injected in some solvent like trioctylphosphineoxide (TOPO) held at a temperature of 300-400 °C. For example when dimethyl cadmium [$Cd(CH_3)_2$] and Se powder were injected in TOPO, CdSe nanoparticles capped with PO_4 groups were obtained. There are, however, other chemicals also which can be used as precursors to obtain high quality particles. It is possible to remove the aliquots at different intervals, as the reaction proceeds, to obtain the particles of different sizes. The particles with high quality and as narrow size distribution as <5% have been achieved by this method.

Although, this is a very good route of synthesizing the nanoparticles, most of the organo-metallic compounds are prohibitively expensive. Besides they are also toxic and difficult to handle. Such synthesis should, therefore, be carried out only under expert guidance.

A wide range of metal oxides and other insulators with wide band gap can be synthesized by chemical precipitation method along with suitable surface passivant, if necessary. Some of the oxides and insulators may be stable and may not agglomerate or coalesce easily.

4.4 LANGMUIR-BLODGETT (LB) METHOD

This technique to transfer organic overlayers at air–liquid interface onto solid substrates is known for nearly 70 years. The technique was developed by two scientists Langmuir and Blodgett and bears their names.

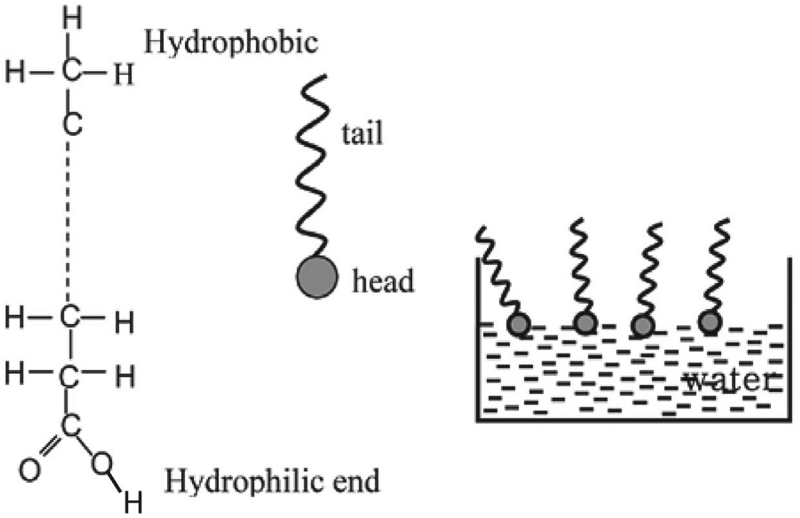


Fig. 4.20: Amphiphilic molecules with hydrophilic and hydrophobic ends to stay with head group immersed in water and tail group in air.

In this technique, one uses amphiphilic long chain molecules like that in fatty acids. An amphiphilic molecule (see Fig. 4.20) has a hydrophilic group (water loving) at one end and a hydrophobic group (water hating) at the other end. As an example consider the molecule of arachidic acid, which has a chemical formula $[CH_3(CH_2)_{18}COOH]$. There are many such long organic chains with general chemical formula $[CH_3(CH_2)_nCOOH]$, where n is a positive integer. In this case $–CH_3$ is hydrophobic and –COOH is hydrophilic in nature.

Usually molecules with $n > 14$ are candidates to form L-B films. This is necessary in order to keep hydrophobic and hydrophilic ends well separated from each other. Figure 4.21 illustrates few examples of different types of molecules which have successfully been used in L-B film deposition.

When such molecules are put in water, the molecules spread themselves on surface of water in such a way that their hydrophilic ends, often called as 'heads' are immersed in water, whereas the hydrophobic ends called as 'tails' remain in air. They are also surface active agents or surfactants. Surfactants are amphiphilic molecules that is an organic chain molecule in which at one end there is a polar, hydrophilic (water loving) and at the other a nonpolar, hydrophobic (water hating) group of atoms.

Using a movable barrier, it is possible to compress these molecules to come closer together to form a 'monolayer' and align the tails. It is, however, necessary that hydrophilic and hydrophobic ends are well separated with $n > 14$. Such monolayers are two dimensionally ordered and can be transferred on some suitable solid substrates like glass, silicon etc. This is done simply by dipping the solid substrate inside the liquid in which ordered organic molecular monolayer is already formed, as shown in Fig. 4.22.

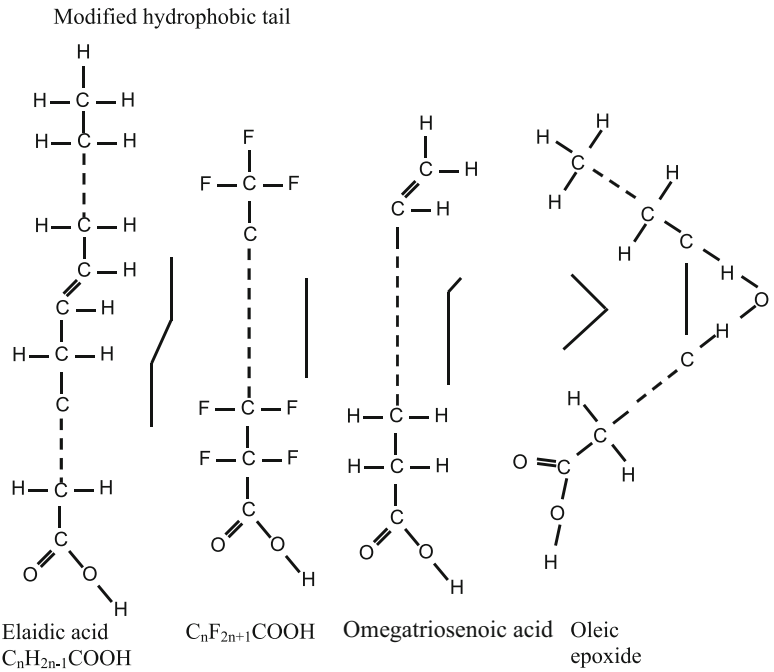


Fig. 4.21: A variety of organic molecules used for L-B thin film deposition.

Depending upon the nature of substrate material i.e. whether hydrophobic or hydrophilic, layers are transferred on the solid substrates. A glass slide when dipped in the solution becomes wet with water. Therefore, while it is withdrawn from the liquid the head groups can be easily attached to glass surface. As a result the whole monolayer gets transferred in a manner as if a carpet is pulled. Now the glass substrate has tail groups, which are hydrophobic on outer side.

Therefore as it is dipped in the liquid again, it acquires a second layer with tail–tail coming closer together and while it is pulled back to air, another monolayer of molecules with head-head groups coming together is pulled. The process of dipping-pulling the substrate can be repeated several times to obtain ordered multilayers of molecules. However to keep ordered layers available on water surface, it is necessary to keep constant pressure on the molecules.

In general there are three types of L-B films with different multilayer sequence, as shown in Fig. 4.23, identified. These are known as X, Y and Z type. Y types of films are most commonly found. Although the layers are ordered, there is only the weak Van der Waals interaction between different layers. In this sense even with large number of layers present, the film preserves its two dimensional properties. Lengths of organic molecules as discussed above are typically 2-5 nm. Thus L-B films themselves are good examples of nanostructured materials.

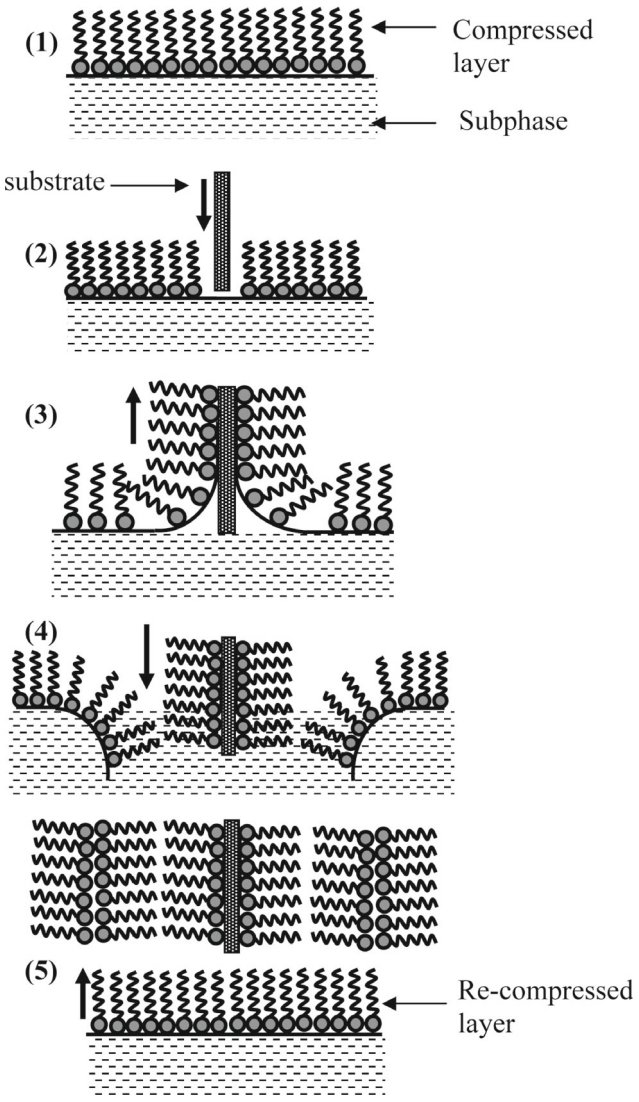


Fig. 4.22: Deposition of LB films by following steps: (1) A monolayer of amphiphilic molecules is formed. (2) A substrate is dipped in the liquid. (3) The substrate is pulled out, during which ordered molecules get attached to the substrate. (4) When the substrate is again dipped, molecules again get deposited as the substrate forming a second layer on the substrate. (5) As the substrate is again pulled out, a thin layer gets deposited. By repeating the procedure a large number of ordered layers can be transferred on a substrate.

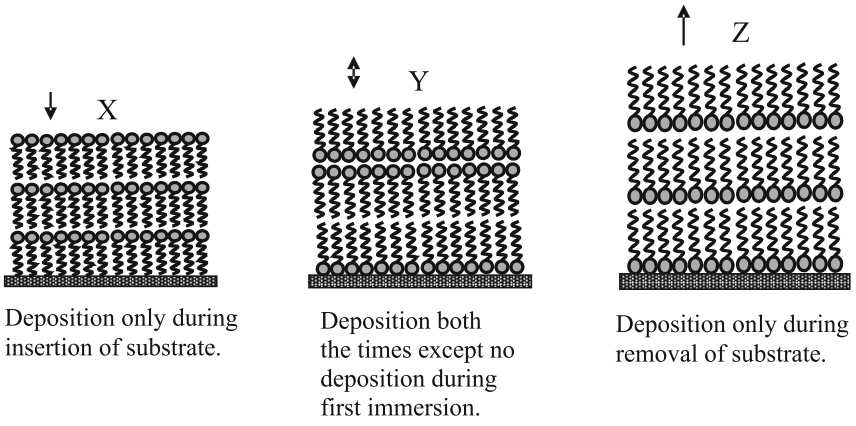


Fig. 4.23: X, Y and Z type L-B films.

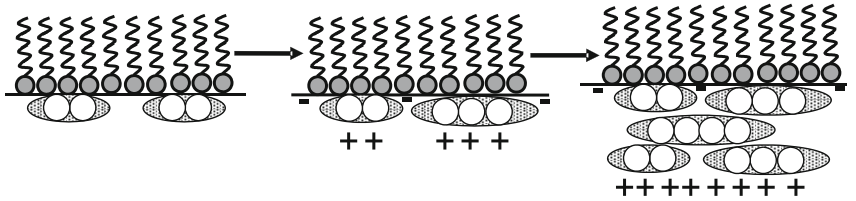


Fig. 4.24: Metal sulphide nanoparticles obtained under the monolayers at water–air interface. The time of hydrogen sulphide treatment increases from top to bottom.

It is also possible to obtain nanoparticles using LB technique. As shown in Fig. 4.24, a metal salt like $CdCl_2$ or $ZnCl_2$ is dissolved in water on surface of which a compressed uniform monolayer (single layer of molecules) of surfactant is spread. When H_2S gas is passed in the solution, CdS or ZnS nanoparticles of few tens of nanometers can be formed. Particles are monodispersed (almost one size) in size. If surfactant molecules are not present, uniform nanoparticles are not formed.

4.5 MICROEMULSIONS

Synthesis of nanoparticles in the cavities produced in microemulsion is a widely used method. Advantage of this method is the biocompatibility and biodegradability of synthesized materials. Biocompatibility is useful in novel applications like drug delivery of nanomaterials and biodegradability is useful to avoid environmental pollution.

Whenever two immiscible liquids are mechanically agitated or stirred together, they are known to form an ‘emulsion’. The tendency of the liquids is such that the

liquid in smaller quantity tries to form small droplets, coagulated droplets or layers so that they are all separated from the rest of the liquid in large quantity (for example droplets of fat in milk). The droplet sizes in emulsions are usually larger than 100 nm upto even few millimeters. Emulsions are usually turbid in appearance.

On the other hand there is another class of immiscible liquids, known as microemulsions which are transparent and the droplets are in the range of ~1 to 100 nm. This is the size needed for the synthesis of nanomaterials.

Box 4.5
Amphiphilic Molecules in Liquids

If amphiphilic molecules are spread in an aqueous solution, they try to stay at air-solution interface with hydrophobic groups in air and hydrophilic groups in the solution (see Fig. 4.25 (a)). Such molecules are known as surfactants (surface active agents). This is similar to what was discussed in L-B film synthesis.

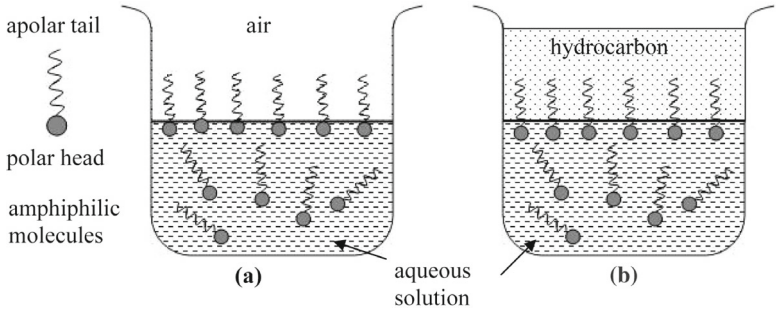


Fig. 4.25: Amphiphilic molecules in aqueous solutions.

Consider now a situation in which a hydrocarbon molecule solution is put in an aqueous medium. As shown in Fig. 4.25 (b), the hydrocarbon solution itself would be separated from aqueous solution and float on it. When surfactant molecules are mixed in large quantity in aqueous solution, they would try to form what are known as 'micelles' and 'inverse-micelles' when aqueous solution is mixed in oil. In micelles, the head groups float in water and tails are inside, whereas tails point outwards in case of inverse micelles.

Box 4.6
Surface Tension of Liquids

Surface tension of a liquid can change if some electrolyte, organic or surfactant solutes are added. General behaviour is shown in Fig. 4.26.

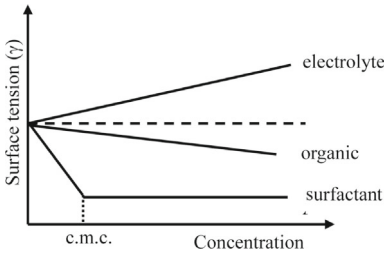


Fig. 4.26: For surfactant molecules γ decreases rapidly upto certain concentration known as critical micelle concentration (c.m.c.).

Microemulsions are stabilized using surfactants (surface stabilized active agents). When an organic liquid or oil (O), water (W) and surfactant (T) are mixed together, under some critical concentration, 'micelles' or 'inverse micelles' are formed, depending upon the concentrations of water and organic liquid. As shown in Fig. 4.27, micelles are formed with excess water and inverse micelles are formed in excess of organic liquid or oil.

It can be seen in Fig. 4.27, micelles have the head groups floating in water, whereas tails and tail group filling the cavity along with organic liquid inside. Reverse is the

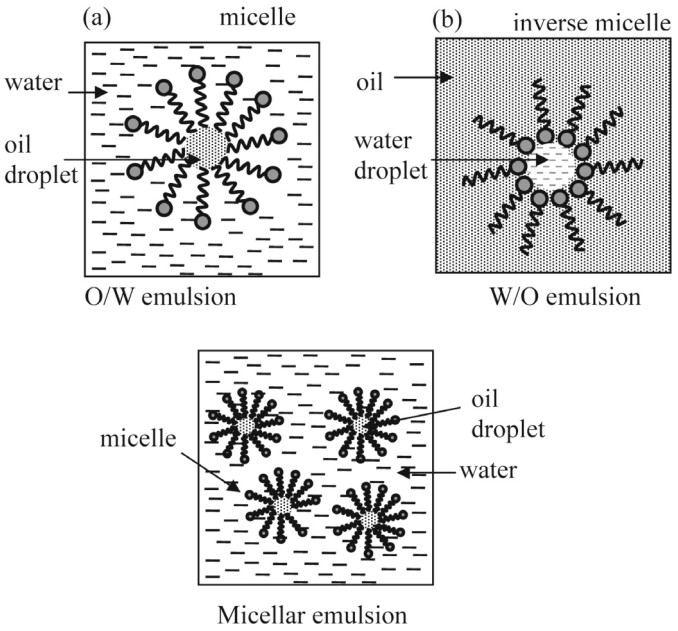


Fig. 4.27: Formation of micelles and inverse micelles.

case for inverse micelles. They can be formed in various shapes as well. In Fig. 4.28 different shapes taken by micelles under different synthesis conditions are illustrated.

The ratio of water (W), oil (O) and surfactant (T) is important to decide which type of micelle will be formed and can be represented in a ternary phase diagram, using a triangle as shown in Fig. 4.29.

Composition can be determined by drawing lines parallel to all three sides of the triangle as shown in Fig. 4.29. Here point *P* denotes 60% water, 26% oil and 14% surfactant.

A modified phase diagram known as 'Winsor Diagram' (see Fig. 4.30 (a)) further denotes the types of phases formed.

There is also another type of phase diagram which shows further details as illustrated in Fig. 4.30 (b).

The critical micelle concentration (CMC) depends upon all W, O and T concentrations as is evident from above diagram. The effect of T is to reduce the surface

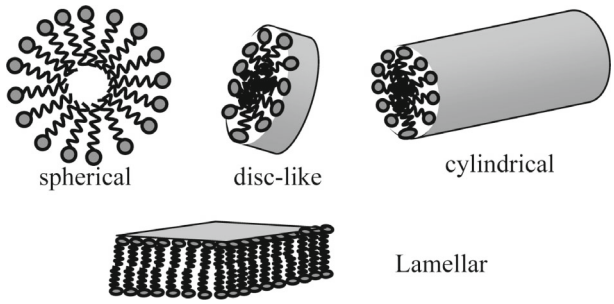
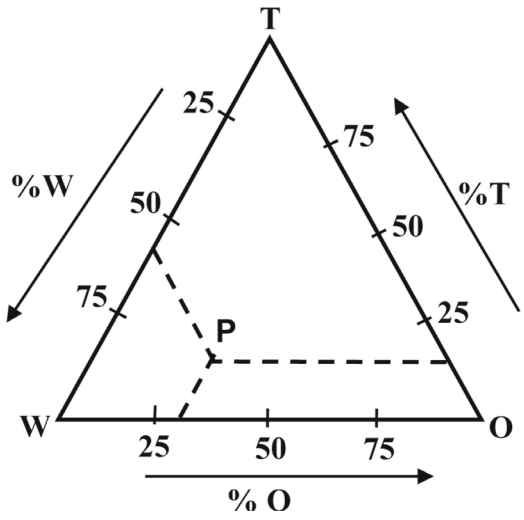


Fig. 4.28: Different shapes of micelles.

Fig. 4.29: Ternary phase diagram of water (W), oil (O) and surfactant (T) mixture.



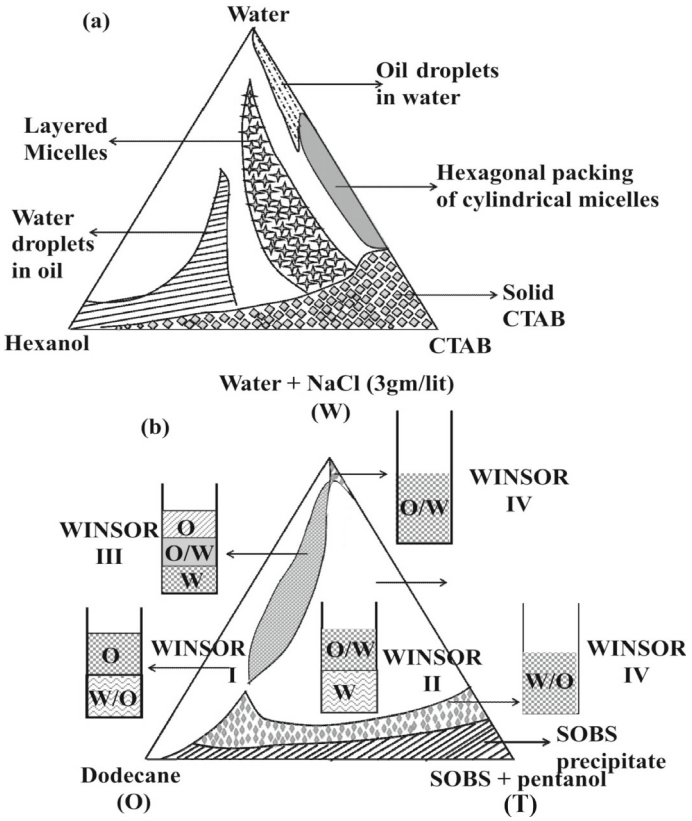


Fig. 4.30: Winsor phase diagram with different phases (a) water-hexanol-centryltrimethylammoniumbromide (CTAB) and (b) water and NaCl-dodecane-paraoctylbenzene sodium sulphate (SOBS) and pentanol.

tension of water dramatically below CMC and remain constant above it, as the organic solvent concentration keeps on increasing. Organic solutes also reduce the surface tension to some small extent. If there are any electrolytes used, they slightly increase the surface tension.

There are four types of surfactants in general:

1. Cationic – For example CTAB, $C_{16}H_{33}N(CH_3)_3{}^+Br^-$
2. Anionic – For example sulphonated compounds with general formula R–$SO_3{}^-Na^+$
   where R is C_nH_{2n+1}.
3. Nonionic – For example R–$(CH_2–CH_2–O)_{20}$–H
4. Amphoteric – Some properties are similar to ionic and some to nonionic surfactants as in betaines.

A large number of nanoparticles (metals, semiconductors and insulators) of cobalt, copper, $CaCO_3$, $BaSO_4$, CdS or ZnS have been synthesized using microemulsions or inverse micelles. As an example, consider the synthesis of cobalt nanoparticles. A reverse miceller solution of water and oil can be stabilized using a monolayer of surfactant like sodium bis (2-ethylhexyl)sulfosuccinate or Na(AOT). The droplet diameter is controlled simply by controlling the amount of water. Two miceller solutions having same diameter of droplets can be formed. Thus one solution should have $Co(AOT)_2$ i.e. cobalt bis(2-ethylhexyl)sulfosuccinate and the other should have sodium tetrahydroborate ($NaBH_4$ i.e. sodium borohydride). When two solutions are mixed together the resultant solution appears clear but the colour changes from pink to black. One can find by electron microscopy or some other analysis that cobalt nanoparticles are formed.

4.6 SOL-GEL METHOD

As the name suggests sol gel involves two types of materials or components, 'sol' and 'gel'. Sol gels are known since the time when M. Ebelman synthesized them in 1845. However it is only since the last one or two decades that considerable interest in it, both in scientific and industrial field, has generated due to realization of the several advantages one gets as compared to some other techniques. First of all sol gel formation is usually a low temperature process. This means less energy consumption and less pollution too. It is therefore not surprising that in the nuclear fuel synthesis it is a desired process. Although sol-gel process generates highly pure, well controlled ceramics it competes with other processes like CVD or metallo-organic vapours derived ceramics. The choice of course depends upon the product of interest, its size, instrumentation available and ease of processing. In some cases sol-gel can be an economical route, provided precursors are not very expensive. Some of the benefits like getting unique materials such as aerogels, zeolites, and ordered porous solids by organic–inorganic hybridization are unique to sol-gel process. It is also possible to synthesize nanoparticles, nanorods or nanotubes using sol-gel technique.

Sols are solid particles in a liquid (see Fig. 4.31). They are thus a subclass of colloids. Gels are nothing but a continuous network of particles with pores filled with liquid (or polymers containing liquid). A sol gel process involves formation of 'sols' in a liquid and then connecting the sol particles (or some subunits capable of forming a porous network) to form a network. By evaporating the liquid, it is possible to obtain powders, thin films or even monolithic solid. Sol gel method is particularly useful to synthesize ceramics or metal oxides although sulphides, borides and nitrides also are possible.

Synthesis of sol-gel in general involves hydrolysis of precursors, condensation followed by polycondensation to form particles, gelation and drying process by various routes as shown in Fig. 4.32. Precursors (starting chemicals) are to be chosen so that they have a tendency to form gels. Both alkoxides or metal salts can be used. Alkoxides have a general formula $M(ROH)_n$, where M is a cation and R is an alcohol

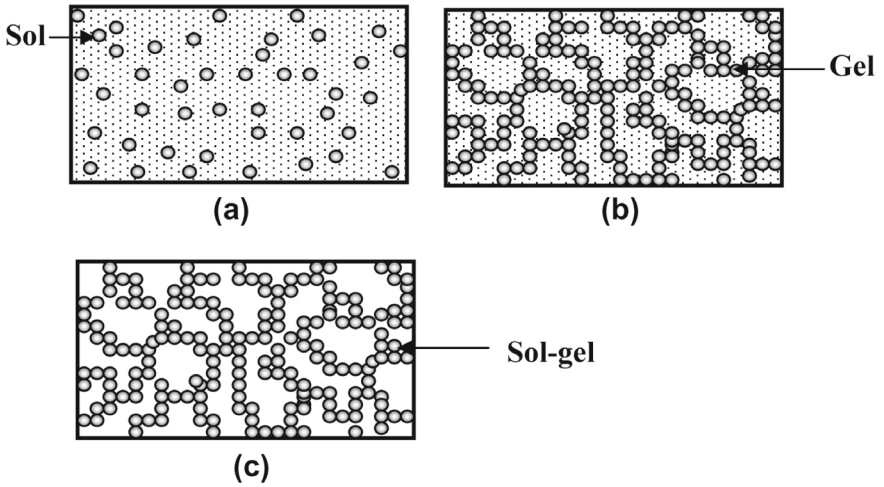


Fig. 4.31: Sol (a), gel (b) and sol-gel (c) monolithic solid.

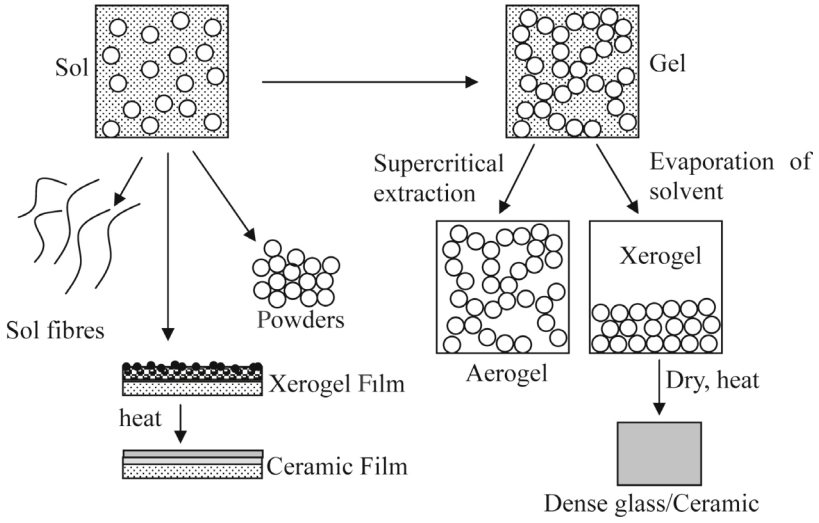


Fig. 4.32: Sol-gel options.

group, *n* is the number of (ROH) groups with each cation. For example (ROH) can be methanol (CH_3OH), ethanol (C_2H_5OH), propanol (C_3H_7OH) etc. bonded to a cation like Al or Si. Salts are denoted as MX, in which M is a cation and X is an anion like in $CdCl_2$, Cd^+ is a cation and Cl^- is an anion.

Although it is not mandatory that only oxides be formed by a sol-gel process, often oxide ceramics are best synthesized by a sol-gel route. For example in silica, SiO_4 group with Si at the center and four oxygen atoms at the apexes of tetrahedron are

very ideal for forming sols with interconnectivity through the corners of tetrahedrons, creating some cavities or pores (see Fig. 3.4 in chapter 3).

Due to its higher electronegativity as compared to metal cations, Si is less susceptible to nucleophilic attacks.

By polycondensation process (i.e. many hydrolyzed units coming together by removal of some atoms from small molecules like OH), sols are nucleated and ultimately solgel is formed.

We shall discuss two special types of sol-gel materials viz. zeolites and aerogels in chapter 10. We shall also see that there are recent methods to combine microemulsion method with sol-gel method to produce some novel materials.

4.7 HYDROTHERMAL AND SOLVOTHERMAL SYNTHESIS

This synthesis method is useful to make a large scale production of nano to micro size particles. In this technique adequate chemical precursors are dissolved in water and placed in vessel made of steel or any other suitable metal which can withstand high temperature typically upto 300 °C and high pressure above 100 bars. The vessel, known as *autoclave*, is usually provided with temperature and pressure control (pressure is generated by the solvents used) as well as measuring gauges as illustrated in Fig. 4.33. When reactions are possible at temperature ~150-160 °C and moderately high pressure is needed, hydrothermal synthesis is successfully carried out in a Teflon reactor placed inside a steel chamber and pressure gauge is eliminated. It is like a 'pressure cooker' used in the kitchen.

It is a very old technique, probably first used by the German scientist Robert Bunsen, way back in 1839 to synthesize crystals of strontium and barium carbonates. He used a thick glass tube and used temperature above 200 °C and pressure more than 100 bars. The technique was later used mostly by geologists and has become popular amongst nanotechnologists due to the advantages like large yield and novel shapes and sizes that can be obtained using this technique.

The technique becomes useful when it is difficult to dissolve the precursors at low temperatures or room temperature. It is also advantageous to use the technique to grow crystalline nanoparticles. The uniformity of shapes and sizes of the nanoparticles also can be achieved by this technique. Various oxide, sulphide, carbonate and tungstate nanoparticles have been synthesized by the hydrothermal synthesis.

Another variation of hydrothermal synthesis technique is known as *forced hydrolysis.* In this case usually dilute solutions (10^{-2} to 10^{-4} M) of inorganic metal salts are used and hydrolysis is carried out at rather higher temperatures than 150 °C.

When the liquids other than water are used as solvents, the term 'solvothermal synthesis' is used to distinguish it from the use of water (hydro).

The technique is very useful to achieve the synthesis of 0-D, 1-D and 2-D nanomaterials. Upscaling is easily achieved to get large quantities of nanomaterials.

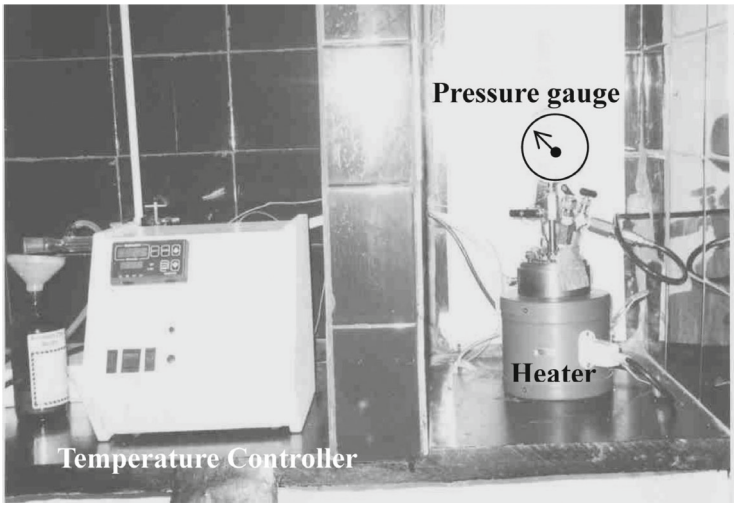


Fig. 4.33: Photograph of an autoclave set up.

4.8 SONOCHEMICAL SYNTHESIS

In this technique the reactivity of the precursors is enhanced by taking the advantage that large amount of energy can be released when bubbles burst in a liquid. Bubbles are formed (see Fig. 4.34) by using ultrasonic waves in a frequency range of ~ 20 kHz – 2 MHz. It can be considered as an alternative method to enhance the chemical reactions in liquids by heating and/or pressurizing.

Although it is not well understood yet as to how nanoparticles can be synthesized using sonochemical method, it is well agreed upon that creation, growth and collapse of bubbles in liquids is most important pathway of causing the reactions. The ultrasonic waves while passing through the liquids create very small bubbles which keep

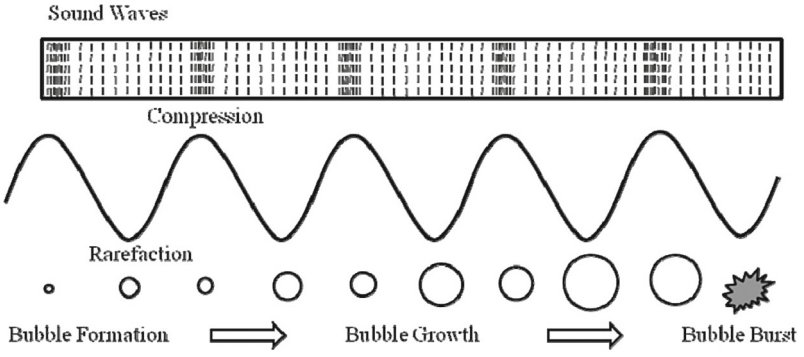


Fig. 4.34: Ultra-sound waves form compressions and rarefactions, shown as a sine wave.

on growing until they reach a critical size and then burst, releasing very high energy to locally reach a temperature of ~5000 °C and a pressure of few hundred times that of atmospheric pressure. For the reaction to occur in the gas phase the solute in the liquid should diffuse to the growing bubble. The reaction can also occur in the liquid phase at exploding bubble where in the interfacial region surrounding the bubble (~200 nm distance) the temperature as high as ~1600 °C can be reached. Typically the size of a bubble can be from ten to few tens of microns. Careful use of the solvents and solutes are very important. Non-volatile liquids would prevent formation of bubbles, which is desired, as only reactants should find their place inside the bubble in the form of vapour. Solvents should be inert and stable to ultrasonic irradiation. Interestingly, the cooling rates also can be as high as 10^{11} °C/s or more. Such a high cooling rate gives rise to amorphous nanoparticles as the atoms do not have sufficient time to reorganize. However amorphous particles can be more active than the crystalline particles of the same size and same material. This can be useful in some fields like catalysis. Various nanoparticles like ZnS, CeO_2 and WO_3 have been synthesized using sonochemical method.

The bubbles after their initiation, grow as shown in the schematic diagram (Fig. 4.34). Bubbles expand progressively in the region of rarefaction and contract at compressions but quickly attain a micrometer size and ultimately burst releasing huge energy. Temperature rises to ~5000 °C and pressure is more than 100 times atmospheric pressure.

4.9 MICROWAVE SYNTHESIS

Use of microwave ovens for heating or cooking food is very common. Its entry in the scientific laboratories took place around 1986 when it was demonstrated by some scientists that rapid, large scale and uniform synthesis of materials is possible even using a domestic microwave oven. Kitchen microwave ovens, however, are not any more considered for the controlled chemical synthesis as there is no good control on stirring, temperature or power which is necessary for scientific equipment. However, due to its several advantages microwave synthesis has been used in research laboratories with equipment capable of controlling various parameters.

Microwaves are a part of electromagnetic spectrum with very long wavelength and frequencies in the range of ~300 to 300,000 MHz. However, only certain frequencies are reserved for domestic and other equipment, rest being used for communication purposes. A microwave has oscillating electric and magnetic fields associated with it which produce nodes and antinodes and correspondingly hot and cold spots in a vessel. This would lead to non-uniformities. Therefore sometimes only a ‘single mode’ is used in which length of the cavity (or reaction vessel) is equal to a single wave only. In a microwave apparatus heating is caused only due to a process in which molecules in a solution try to orient their dipoles appropriately to align themselves in the direction of the electric field. In the process they too oscillate, generating heat in the medium. Advantage in this case is that the external energy is not wasted in

heating of the vessel. Moreover at 2.45 GHz, a commonly used frequency for the microwave oven, only 1.0×10^{-5} eV or ~1 J mol^{-1} allows the necessary heating. Thus the use of microwave is a highly energy efficient process and requires small time for the reactions to take place.

Several types of oxide, sulphide and other nanoparticles have been synthesized to obtain various shapes and sizes. The reaction time is greatly reduced and the products are uniform in size and shape.

4.10 SYNTHESIS USING MICRO-REACTOR OR LAB-ON-CHIP

Micro-reactor or lab-on-chip is a very convenient method of synthesizing nanoparticles in small quantities. Basically, very narrow channels (less than about 100 μm upto few tens of nm in width and depth) are made in some suitable substrates like glass, silicon or polymers like poly-dimethylsiloxane (PDMS) using lithography techniques. Similar to an electronic circuit in a semiconductor chip, these channels make some circuit where the fluids can mix. There can be some mixer regions where stirring takes place with the help of magnetic or some other actuation. There can be some valves to control the flow of liquids. The channels can be of short or long length depending upon the requirements and are designed to suit a particular requirement. The size of the whole reactor can be a few cm^2. The liquids may be aqueous or non-aqueous and suitable reactor will have to be chosen according to the reactions to be carried out. It is also possible to heat some of the reactors to enhance the rates of reaction. The liquids should not react with the reactor material or percolate inside its body. The liquids are injected inside the channels using syringe pumps. Figure 4.34 illustrates few simple designs of a microreactor set up without complicated valves, mixers and other components (Fig. 4.35).

Advantage of synthesizing nanomaterials in microreactors is that reactions can be carried out in a very short time using small amounts of reactants. This is advantageous when expensive or toxic reactants are to be used. Due to small amounts involved, the risk of pollution is minimized. Short synthesis time enables many reactions to be carried out in a short time. Therefore, optimization of reaction parameters can be done very quickly. However, due to small channel size, the fluid flow in channels and in large glass flask reactor may differ. However, using parallel processing one can increase the quantity of the product to be obtained in microreactors. One, however, has to be careful that if the particles grow to large size they may clog the channels. Cleaning of the channels also can pose problems. However disposable polymer-based inexpensive microreactors can be a good solution to avoid the cross contaminations.

There are many reports now which show that TiO_2, ZnS, CdSe, Au, Ag etc. nanoparticles with narrow size distribution have been achieved using microreactors. Moreover doping of nanoparticles also is possible in microreactors. In short, whatever

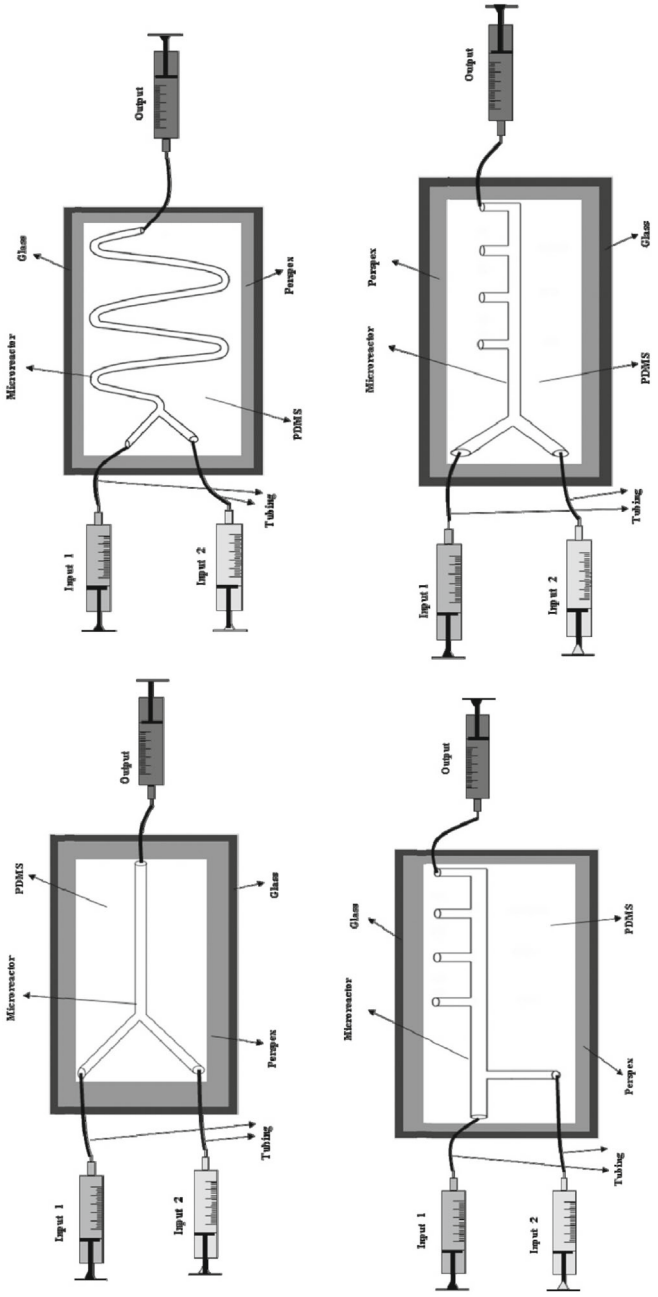


Fig. 4.35: Schematic of microreactor set up.

the synthesis is carried out in a chemistry laboratory can be carried out in the lab-on-chip.

FURTHER READING

Doped II-VI Semiconductor Nanoparticles by S.K. Kulkarni. Encyclopedia of Nanoscience and Nanostructured Materials, **2**: 537 (2004).

Introduction to Sol-Gel Processing by Alan C. Pierre. Kluwer Academic Publishers, Boston (1998).

Sol-Gel Science: The Physics and Chemistry of Sol-Gel Processing by C. Jeffrey Brinker, J. W. Scherer. Academic Press. Boston (1990).

Brinker, J.W. Scherer. Academic Press, Boston (1990).

Fundamental Principles of Sol-Gel Technology by R.W. Jones. The Institute of Metals (1989).

Basic Principles of Colloid Science by D.H. Everett. Royal Society of Chemistry (1988).

Monodispersed Metal (hydrous) Oxides—A Fascinating Field of Colloid Science by E.J. Matijevic. *Acc. Chem. Res.*, **14**: 22-29 (1981).

Chapter 5
Synthesis of Nanomaterials—III (Green Synthesis/Biological Methods)

5.1 INTRODUCTION

In Chapters 3 and 4 synthesis of nanomaterials using physical and chemical methods was discussed. These methods although efficient often use expensive or toxic precursors/solvents (except water). In some cases there is an emission of polluting gases or effluents, which affects environment. This in turn creates problems to human, animals, plants and agriculture. Therefore there is a need of Green Synthesis or ecofriendly methods to obtain nanomaterials. Way back in 1959, in his very famous speech, before the scientists of American Physical Society, Nobel Laureate Richard Feynman asked the scientists to derive the inspiration from *Mother Nature* to make the things smaller and see the advantages of making things smaller. Indeed the biological world, animal kingdom and plants make optimum use of materials and space. Nature indeed makes use of small spaces and corresponding confinement to synthesize inorganic materials or minerals abundantly found in earth's crust. It uses insoluble, complex organic molecules as a reactor in which nucleation and growth of complex and hierarchical structures of inorganic materials takes place by reactions of organic soluble molecules. When we think of biological world, we normally think of delicate, temperature sensitive, carboneous or organic materials like leaves, roots, stems, cells, tissues and skin. Inorganic materials are also produced in biological systems. Bones, teeth, shells, nanomagnets inside the bodies of some bacteria and birds are some examples. Inorganic materials inside organic matter or organisms are known as biocomposites or biominerals. A variety of mechanically strong or weak, rigid or flexible, porous or nonporous, thick or thin materials either organic or inorganic in small or large quantities are abundantly produced in contact with live cells. These materials exhibit a wide variety in their functions like providing support to body, allow body movements and in general carry out various essential body functions. Table 5.1 gives a list of some biominerals produced or observed in the biological systems. The list is not at all complete and many more minerals are observed to be synthesized in nature under different environmental conditions.

6.2 MECHANISM OF SELF ASSEMBLY

It is important to realize that the self assembly involves weak to strong forces and nanoscopic to gigantic structures in one, two or three dimensions. As mentioned in the introduction, the self assembly can be a result of very weak forces like van der Waals interaction, hydrogen bonds, static charges, magnetic interactions and so on. The driving force behind the self assembly has been recognized as due to attempt of any system to go to the lowest energy state. Ability of a system to go to a well ordered low energy state depends upon the availability of units of same size and shape. Molecules with definite shape, number of atoms and therefore size, already in the state of low energy are good candidates for the self assembly. It should be remembered that atoms in such molecules or building blocks of self assembly themselves are held together by stronger forces.

When the building blocks (one, two or more types of building blocks also can form self assemblies) are available for self assembly, state of minimum energy may be attained spontaneously in the absence of any external force. However self assembly may occur in the presence of an external driving force like temperature, pressure, magnetic field and so on. These two types viz. assembly in the absence and presence of an external driving force are known as *static* and *dynamic* assemblies respectively (see Fig. 6.1). Static assembly is realized when system achieves minimum energy state and can stay there unless it is subjected to strong external forces. On the other hand, dynamic self assembly involves constant influence of external force from the ambient. If the energy intake from the ambient stops, the self assembly can leave the state of organized structure and de-assemble. Formation of ordered crystalline structure from a melt can be considered as an example of static self assembly. Thin films formed by L-B technique (Chapter 4) or dip pen lithography technique (to be discussed in Chapter 11) also are examples of static self assembly. The living animals or plants form good examples of dynamic self assembly. As soon as the supply of food, proper temperature and air pressure discontinue, the animals and plants disintegrate.

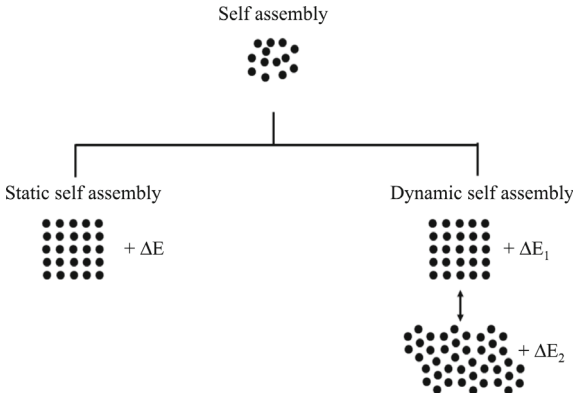


Fig. 6.1: Two types of self assemblies.

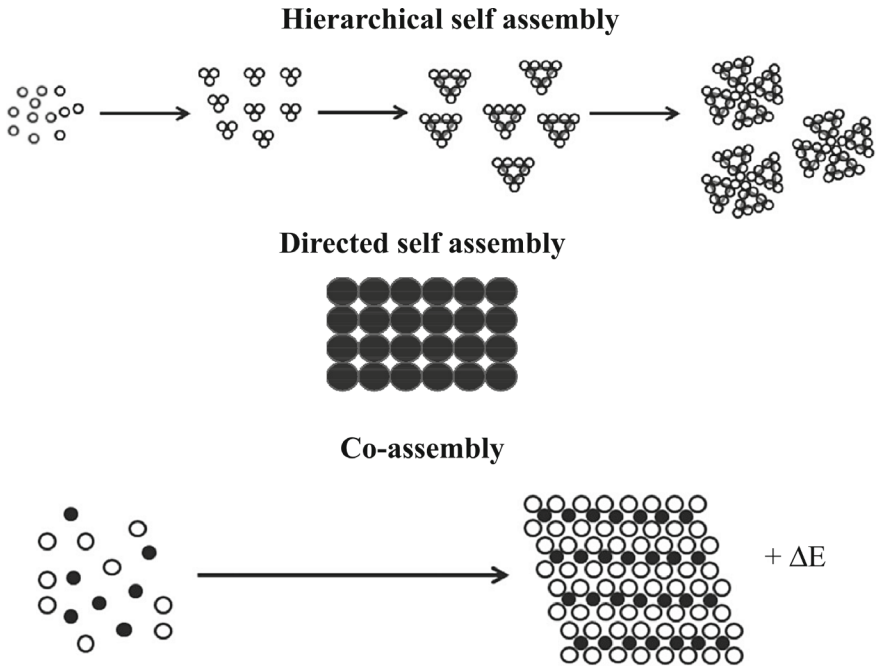


Fig. 6.2: Hierarchical and co-assembly.

The static and dynamic assemblies can be further divided into '*hierarchical self assembly*', '*directed self assembly*' and '*co-assembly*' as illustrated schematically in Fig. 6.2.

Hierarchical self assembly is characterized by small range, medium range and long range interactions of one type of a building block.

Directed self assembly occurs when the building blocks occupy the pre-designed places like some portions of a lithographically patterned substrate, pores in membranes or spaces between ordered particles.

Co-assembly, as the name suggests, can be formed with two or more types of blocks which can fit into each other.

We shall discuss in the following section some examples of self assembly. Evaporation or biological templates like DNA or S-layers are useful for self assembly. In most of these cases, it is possible to place the nanoparticles at some well determined location of polymer chain, organic molecule or template through specific bonding. When well structured templates like S-layers or DNA are used, direct patterning of nanoparticles becomes feasible. These are the examples of inorganic (nanoparticles)-organic molecules or inorganic-biomolecules assemblies. However it is also possible to have assemblies of purely inorganic particles. The driving force for such assemblies is often quite different. We shall discuss now little more details of various assemblies.

6.3 SOME EXAMPLES OF INORGANIC SELF ASSEMBLY

Self Assembly of a variety of rod or disc like molecules gives rise to well known 'Liquid Crystals' widely used in display panels. In the following few examples of solid nanoparticles, assembled regularly like a crystal, with precise spacing in two or three dimensions, are given. These assemblies have an advantage that they can be controlled at molecular level. Precise assemblies without disorder and defects are most likely to occur when the components (coassembly) of Self Assembly are of uniform size and shape. They should not flocculate or form aggregates by strong ionic or covalent bonding but held together by weak interactions like van der Waals, hydrogen bond, electrostatic and dipole-dipole. It is common to achieve 0-D, 1-D, 2-D and 3-D (superlattices) self assemblies of inorganic solid nano particles. Figure 6.3 schematically shows these possibilities. Anisotropic particle and polyhedral, monodispersed particles of same shape also condense on different surfaces with precisely controlled synthesis and environment.

6.3.1 Self Assembly of Nanoparticles Using Organic Molecules

Self assembly of nanoparticles is a very useful way of obtaining nanoparticle assemblies at low cost in controlled fashion. Long chain organic molecules as surfactants (or even some specifically designed short chain molecules), decomposition of surfactant micelles, destabilization of solvent and confinement in emulsion droplet are some of the ways in which Self Assembly is initiated in liquids. When the liquid droplets are placed on suitable substrates at controlled temperature, surrounding atmosphere and mechanical stability (no vibrations), liquid starts evaporating and nanoparticles of ~1 to 100 nm come closer together to form two dimensional or three dimensional assemblies over hundreds of nanometer range. The process may take few minutes to

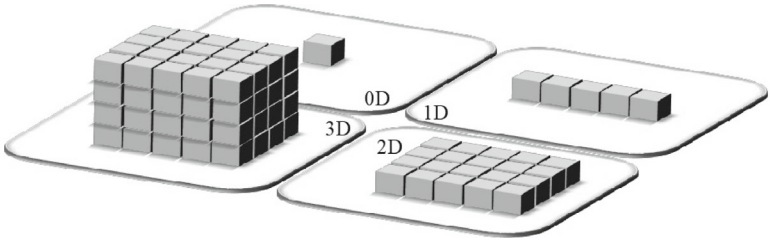


Fig. 6.3: A schematic presentation of 0-D, 1-D, 2-D and 3-D assemblies of nanoparticles on solid surfaces.

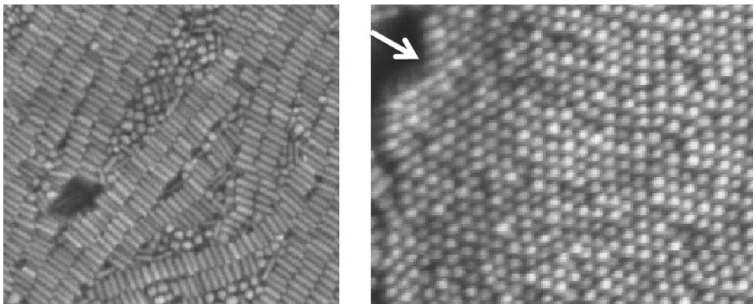


Fig. 6.4: Field Emission Scanning Electron Microscopy (FESEM) Self Assembly Images of 38 nm × 9 nm Gold Nanorods (a) in plane and (b) as vertical assembly, a side view. Vertical alignment is indicated by an arrow in the left hand top corner.

few hours depending on the solvent. Self Assembly is a bottom-up process that takes place via nucleation and growth. The growth may be homogeneous or heterogeneous.

Figure 6.4 shows an example of Self Assembled gold nanorods on ordinary, clean glass plates used as substrates. Nanorods of gold are synthesized by chemical synthesis process outlined in Chapter 15. Assembly is achieved via liquid drop casting and slow evaporation.

Preformed inorganic nanoparticles can be assembled on solid substrates through some organic molecules adsorbed on their surfaces. For example CdS nanoparticles functionalized with carboxylic (-COO^-) group can be transferred (Fig. 6.5 (a)) to aluminium thin films. Dithiols adsorbed on metal surfaces also could adsorb CdS nanoparticles (Fig. 6.5 (b)) to form layers of them. Silver particles (Fig. 6.5 (c)) have been adsorbed on oxidized aluminium layers using bi-functional molecule such as carboxylthiophenol. Molecules bind to aluminium oxide layer through carboxylic group and thiol attaches to silver particles.

Using a two-phase reaction alkanethiol or alkylamine capped gold, silver, palladium etc. nanoparticles have been self assembled. Here chemical reaction takes place in an aqueous medium. The particles are then transferred into an organic solvent and drop casted on an appropriate solid substrate. Solvent is allowed to evaporate which leaves self assembled layer. Preformed inorganic particles of materials like silica (SiO_2), titania (TiO_2). Polymer beads or latexes are able to organize themselves just by sedimentation also. But they need to have very uniform size. As illustrated in Fig. 6.6, organization of SiO_2 particles is quite clear. The silica particles can be formed by a sol-gel route. The particles synthesized in aqueous medium are simply allowed to evaporate from the solution drop placed on a glass substrate. After some time the particles self assemble due to weak van der Waals interaction amongst the particles. The driving force is the capillary force. Minimization of surface energy takes place by forming a hexagonal network. The uniform size of the particles is helpful to make an ordered two dimensional network of particles.

In addition to spherical and anisotropic particles self assembly of polyhedrons has been achieved. Metal Organic Frame (MOF) or Covalent Organic Frame (COF)

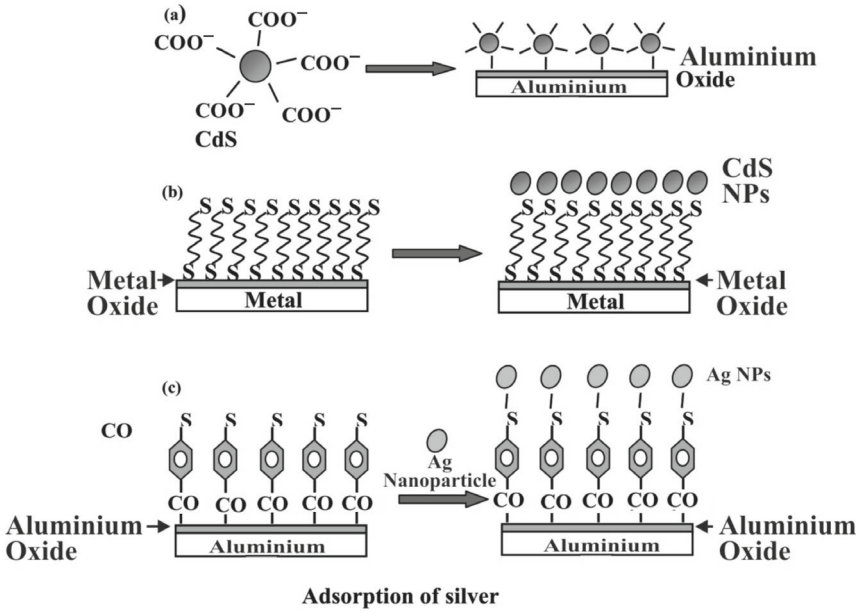
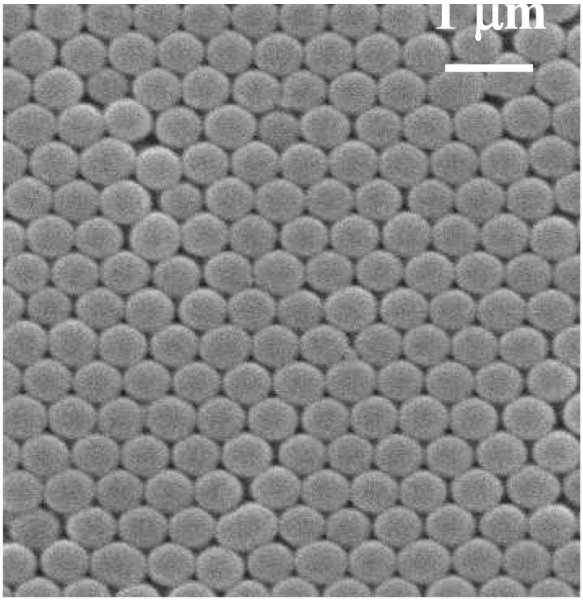


Fig. 6.5: Self assembly of nanoparticles.

Fig. 6.6: SEM image of self assembled silica (SiO_2 particles) on glass slide.



work which have central metal or non-metal around which organic framework is formed giving rise to polyhedral units or particles. Materials are highly porous. These materials also are found to exhibit 0-D, 1-D, 2-D and 3-D self assemblies.

In general, use of electric, magnetic and electromagnetic fields while evaporating solutions to assemble polar and magnetic particles can be made. This helps the nanoparticles to align themselves in the direction of the field.

6.3.2 Self Assembly in Biological Systems

There are many examples of self assembly in the biological systems like S-layers, proteins or DNA. When organized arrays of inorganic crystals are embedded in biological systems they are often referred to as biomineralized systems. Here we shall discuss only the ordered self assembled layers as examples of biomineralization.

Magnetotactic bacteria are small bacteria, with ~35 to 120 nm sized permanent magnets inside them. The magnets are of either iron sulphide (Fe_3S_4-greigite) or iron oxide (Fe_3O_4-magnetite). Such magnets make a chain of nanomagnets. The size 35 to 120 nm is quite critical and smaller or bigger magnets would not have served the purpose for which these magnets are used. The magnets smaller than 35 nm cannot have permanent magnetism at ambient temperature and those larger than 120 nm would have reduced magnetism due to multidomain formation above this size. Only those between 35 to 120 nm size are single domain permanent magnets, the chain of which is useful for navigation of bacteria. These bacteria live in mud, marshy areas, ponds or sea and prefer anaerobic condition. If by any chance they come to the surface of water or soil, they navigate downwards making use of their magnets aligning in the earth's magnetic field direction. Earth's magnetic field has a dip in the north and south hemispheres which helps bacteria to seek downwards direction. Even some birds have ordered nanomagnets which they sometimes use for navigation purpose.

A very common self assembly in biological systems is S-layers. They are one of the simplest kind of biomembrane evolved during billions of years and is invariably a part of cell envelope of prokaryotic organisms, with just a few exceptions. They are two dimensional, crystalline single proteins or glycoprotein monomers organized in hexagonal, oblique or square lattices, as illustrated schematically in Fig. 6.7.

These lattices have ordered pores. The periodicity of pores can vary, depending upon the protein, from 3 to 35 nm. In some cases pores ~2 to 8 nm have also been observed.

Such S-layers after extraction from bacterial cells have been transferred on some metallic substrates (or grids). When treated with cadmium salt and subsequently with Na_2S, ordered arrays of CdS nanoparticles could be formed. This is confirmed using electron microscopy. S-layers are reported to have been used to assemble Au, Pt, Fe and Ni metal nanoparticles. In general, S-layers extracted from the biological cells can be directly used to deposit nanoparticles from liquid or vapour phase as shown in Fig. 6.8.

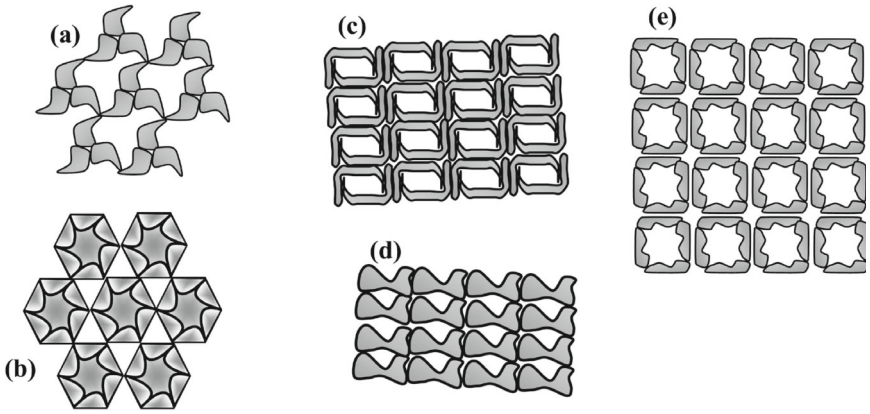


Fig. 6.7: Different types of S-layer lattice.

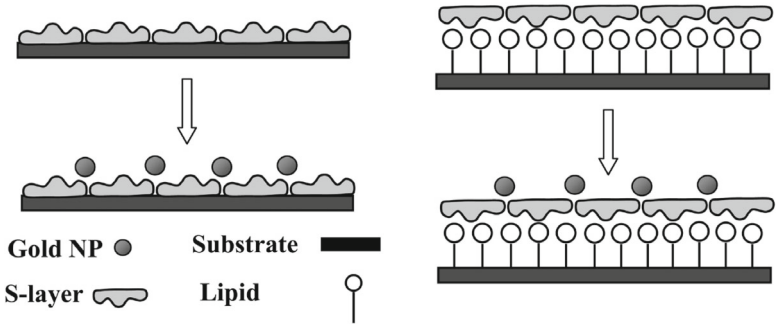


Fig. 6.8: Assembly of CdS nanoparticles using S-layer.

It was discussed in Chapter 5 that ferritins are protein colloids of 12 nm size found in all animals. Ferritins have cavities ~6-8 nm in size filled with iron oxide. It is possible to remove iron oxide and replace it with metal or other nanoparticles. Further it is possible to make a two dimensional array of ferritins in solution. For example ferritin solution in NaCl and phosphate at ~5.8 pH can be filled in a trough. Chloroform containing dichloroacetic acid can be used to dissolve poly-1-benzal-L-histidine (PBLH) and spread over ferritin solution in the trough. After about two hours the solution can be heated at 38 °C for one hour and cooled back to room temperature. This produces ordered layer of ferritin at liquid-air interface. The layer can be transferred on silicon substrate by dipping it in the solution. An ordered layer of ferritins is transferred on the surface. By heating the silicon substrate for one hour in nitrogen ambient at 500 °C, protein from ferritin is completely removed, leaving an ordered layer of iron oxide or whatever nanoparticles were filled in ferritin. There are several reports of two dimensional ordered layers being transferred by this route of Co, CoPt, FePt. The major application of this type of layers is in high density

magnetic data storage systems. It is speculated that CoPt particles in this manner will be able to have as large as 1550 Gbit/cm^2 per particle. This is quite large compared to current ~11 Gbit/cm^2.

DNA (deoxyribonucleic acid) is a long helical molecule and was discussed in Chapter 5. It has a large aspect ratio (i.e. ratio of length to diameter is large) and acts like a long one dimensional template in its simplest form. Its four nucleotide bases viz. quinine (Q), cytosine (C), adenine (A) and thymine (T) can form a rich variety of sequences and structures. Thus circular, square, branched and many more long (few micrometre) or short (few nanometre) DNA templates are possible. Besides planer geometry, they can adopt even three dimensional structures. As DNA has alternate sugar and phosphate groups on its strands, it is possible to anchor metal, semiconductor or oxide particles by different bonding on DNA to have assembly of particles.

6.3.3 Self Assembly on Semiconductor Surfaces

It is possible to spontaneously create the quantum dots for example of germanium (Ge) on silicon (Si) or indium arsenide (InAs) on gallium aresenide (GaAs) indium nitride (InN) on gallium nitride (GaN) and so on. This is possible on high quality single crystal faces and in highly controlled environment (ultra high vacuum) and specific temperature. The origin of self assembly is strain induced. As an example, germanium and silicon have only 4% lattice mismatch. Therefore Ge can be deposited epitaxially on Si single crystal upto 3-4 monolayers. Although grown (hetero)epitaxially, the layers of deposited Ge are highly strained (coherently i.e. without any defects or dislocations). When further deposition takes place, the lattice strain caused by depositing Ge on Si with different lattice constants cannot be accommodated.

This results in spontaneous formation of nanosized islands or quantum dots. However the temperature of the substrate has to be >350 °C during deposition or post-deposition annealing is required. Figure 6.9 schematically illustrates the growth mechanism as well as an electron microscopy image of germanium islands on Si (111) surface. The size of the islands depends upon the growth temperature as well as the substrate plane on which it grows.

Self assembly thus has a very rich variety of practical examples. Many other self assemblies can also be designed and fabricated. Such assemblies have a great potential in molecular information technology as it is possible to store information and process information. It also is useful for nanofabrication. If complex arrangements of self assembly can be made like in biological processes, compact devices can also be realized using self assembly. One such example given by Lehn is the brain which is self assembled and wired by self organization to different organs. It is able to store information, process and also has control over different body parts, all through self assembly or organization.

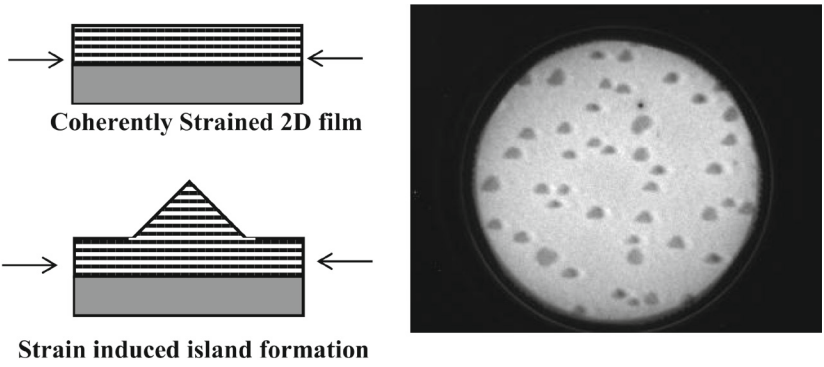


Fig. 6.9: Growth mechanism of Ge on Si and photograph showing island formation (field of view 10 μm).

FURTHER READING

Javier Fonseca, Lingxin Meng, Inhar Imaz and Daniel Maspoch. Self-assembly of colloidal metal–organic framework (MOF) particles. *Chem. Soc. Rev.*, **52:** 2528 (2023).

Materials Today, special issue on Nanofabrication by Self Assembly, Vol. 12 (May 2009).

Core Concepts in Supramolecular Chemistry and Nanochemistry by Jonathan W. Steed, David R. Turner and Karl J. Wallace. John Wiley & Sons Ltd. (2007).

Introduction to Supramolecular Chemistry by Helena Dodziuk. Springer (2002).

Supramolecular Chemistry: Concepts and Perspectives by J.–M. Lehn. VCH Weinheim, Germany (1995).